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Permissioned Distributed Ledger (PDL); Research and Innovation Landscape

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Foreword

This Group Report (GR) has been produced by ETSI Industry Specification Group (ISG) Permissioned Distributed Ledger (PDL).

Modal verbs terminology

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Executive summary

The present document gives an overview of relevant research and standardization projects in the field of permissioned distributed ledgers and Blockchain, in particular identifying projects from the EU's Horizon 2020 programme, the US' National Science Foundation's programme, national programmes in the UK and Germany, in addition to worldwide standardization activities in this field.

Introduction

Clause 4 introduces research and innovation programmes relevant to permissioned distributed ledgers, in particular the European Union's Framework Programme Horizon 2020 and its primary information repository, the Community Research and Development Information Service (CORDIS). Clause 5.1 gives a general overview of the EU's Horizon 2020 Framework Programme, its successor Horizon Europe and the Connecting Europe Facility (CEF), while clause 5.2 focuses on the legal framework for standardization and an overview of blockchain-related standardization activities.

Clause 6 provides an overview of public research and innovation programmes and grants related to PDL, covering the EU in more detail, and in addition national programmes in Germany, the UK and the US. Clause 7 emphasizes the importance of PDL standardization. Annex A identifies in detail those EU Horizon 2020 projects that are related to DLT and Blockchain, sorted by project name, while Annex B identifies for those same projects the related Horizon 2020 research topics and thematic priorities. Finally, Annex C lists US National Science Foundation (NSF) projects related to DLT and Blockchain.

1 Scope

The present document shows the current research and innovation programmes related to permissioned distributed ledgers, distributed digital ledger technologies and blockchain with the goal of identifying advanced technologies and innovative research results relevant or essential to PDL standardization.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI GR PDL 001 (V1.1.1): "Permissioned Distributed Ledger (PDL); Landscape of Standards and Technologies".

NOTE: Available at https://www.etsi.org/deliver/etsi_gr/PDL/001_099/001/01.01.01_60/gr_PDL001v010101p.pdf.

[i.2] European Commission CORDIS: "CORDIS: Projects and Results".

NOTE: Available at https://cordis.europa.eu/projects/en.

[i.3] European Commission, DG for Communication Networks, Content and Technology Call for tenders /(25-11-2020): "(European Commission Call for tenders CNECT/2020/OP/0055),EU Blockchain Pre-commercial Procurement, Open procedure, Tender Specifications".

NOTE: Available at https://etendering.ted.europa.eu/cft/cft-document.html?docId=81917; https://etendering.ted.europa.eu/document/document-file-download.html?docFileId=95923.

[i.4] European Union Regulation (1025/2012).

NOTE: Available at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32012R1025&from=EN

[i.5] ETSI Annual Report 2019, published in 2020.

NOTE: Available at https://www.etsi.org/e-brochure/Annual-Reports/AR-202012/mobile/index.html.

[i.6] EC Rolling Plan for ICT Standardization, Published April 2021.

NOTE: Available at https://joinup.ec.europa.eu/collection/rolling-plan-ict-standardisation/rolling-plan-2021 , the Rolling Plan addresses technology areas in need of ICT standards and explores the role that standards and technical specifications can play in achieving the policy objectives. It reaches out to both European Standardisation Organisations (ESOs) -ETSI, CEN and CENELEC and aforementioned global standard development bodies that can respond to the proposed actions and support the respective policy objectives with standardisation deliverables.

[i.7] European Commission Horizon 2020 Work Programme /(2020-06-17): "EN Horizon 2020 Work Programme 2018-2020, 5.i. Information and Communication Technologies".

NOTE: Available at https://ec.europa.eu/research/participants/data/ref/h2020/wp/2018-2020/main/h2020-wp1820-leit-ict_en.pdf.

[i.8] European Commission European Research Council 2021 Work Programme /(2021-02-22): "(European Commission Decision C(2021) 930 of 22/02/2021), ERC Work Programme 2021, European Research Council".

NOTE: Available at https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2021/wp_horizon-erc-2021_en.pdf.

[i.9] Federal Ministry for Economic Affairs and Energy, Federal Ministry of Finance/(2019-09-18): "Blockchain Strategy of the Federal Government, We Set Out the Course for the Token Economy".

NOTE: Available at https://www.bmwi.de/Redaktion/EN/Publikationen/Digitale-Welt/blockchain-strategy.pdf.

[i.10] T. Faisal et el.: "Automated Quality of Service Monitoring for 5G and Beyond Using Distributed Ledgers", IWQoS 2021.

[i.11] T. Faisal et el.: "How to Request Network Resources Just-in-Time using Smart Contracts", ICBC 2021.

[i.12] T. Faisal et el.: "AJIT: Accountable Just-in-Time Network Resource Allocation with Smart Contracts", MobiArch 2020.

[i.13] T. Maksymyuk, M. Volosin, G. Bugar, D. Horvath, J. Gazda, M. Klymash and Mischa Dohler: "Blockchain-Based Comprehensive Network Management in 5G and Beyond," IEEE Communications Magazine, accepted 23 August 2020.

[i.14] C. Ngubo, Mischa Dohler: "WiFi Dependent Consensus Mechanism For Constrained Devices using Blockchain Technology," IEEE Access (Journal), accepted 24 July 2020.

[i.15] R. Pirmagomedov, et al, and Mischa Dohler: "Applying Blockchain Technology for Users Incentivization in mmWave-based Mesh Networks," IEEE Access (Journal), accepted 17 Feb 2020.

[i.16] Setting standards for the future.

NOTE: Available at https://www.sofie-iot.eu/news/setting-standards-for-the-future.

[i.17] 5GZorro.

NOTE: Available at https://www.5gzorro.eu/our-objectives/.

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

5G-CORAL	5G Convergent Virtualized Radio Access Network Living at the Edge (H2020 project)
AEGIS	Advanced Big Data Value Chain for Public Safety and Personal Security (H2020 project)
AGILE	Adoptive Gateways for dIverse muLtiple Environments
AI	Artificial Intelligence
AJIT	Accountable Just-in-Time Network Resource Allocation (H2020 project)
ANITA	Advanced tools for fighting oNline llegal TrAfficking (H2020 project)
API	Applications Programming Interface
ATARCA	Accounting Technologies for Anti-Rival Coordination and Allocation (H2020 project)
ATM	Air Traffic Management
BBI-CSA	Bio-Based Industries - Coordination and Support Action
BCDMA	Blockchain Division Multiple Access
BDLT	Blockchain and DLT
BDTI	Big Data Test Infrastructure
BMBF	German Federal Ministry of Education and Research (Bundesministerium für Bildung und Forschung)
BMWi	German Federal Ministry for Economic Affairs and Energy (Bundesministerium für Wirtschaft und Energie)
CAPS	Collective Awareness Platforms
CBDC	Central Bank Digital Currency
CCF	Connecting Capability Fund
CEF	Connecting Europe Facility
CEITER	Cross-border Educational Innovation thru Technology-Enhanced Research
CEN	European Committee for Standardization (Comité Européen de Normalisation)
CENELEC	European Committee for Electrotechnical Standardization (Comité Européen de Normalisation Électrotechnique)
COG-LO	COGnitive LOgistics Operations through secure, dynamic and ad-hoc collaborative networks (H2020 project)
COPA	COllaborative Platform for trAnsmedia storytelling and cross channel distribution of EUROPE
CORDIS	Community Research and Development Information Service
CPN	Content Personalization Network
CREST	Fighting Crime and TerroRism with an IoT-enabled Autonomous Platform based on an Ecosystem of Advanced IntelligEnce, Operations, and InveStigation Technologies (H2020 project)
CRUE	H2020 project
CSA	Coordination and Support Action
CSA-LS	Coordination and Support Action Lump Sum
CUREX	seCUre and pRivate hEalth data eXchange (H2020 project)
DFS	Democratized Financial Services (H2020 project)
DG CNECT	Directorate-General for Communications NEtworks, Content and Technology
DLT	Distributed Ledger Technology
DSISCALE	Digital Social Innovation in Europe (H2020 project)
DSO	Distribution System Operator (of electricity)
EBSI	European Blockchain Services Infrastructure
EBSIS	Event Based Systems (H2020 project)
EC	European Commission
ECHO	European network of Cybersecurity centres and competence Hub for innovation and Operations (H2020 project)
ECSEL-RIA	ECSEL Research and Innovation Action
ECSO	European Cyber Security Organisation
EECTI	Spanish National Strategy for Science, Technology and Innovation
EGNSS	European Global Navigation Satellite System
EHR	Electronic Health Record
EIC	European Innovation Council
eIDAS	Electronic IDentification, Authentication and trust Services
EIT	European Institute of Innovation and Technology
ENIT	Energy IT Systems

NOTE: ENIT is a spinoff company of Fraunhofer Society (German research society).

ENVRI PLUS	Environmental Research Infrastructures Providing Shared Solutions for Science and Society (H2020 project)
EPES	Electrical Power and Energy System
EPSRC	Engineering and Physical Sciences Research Council
ERA	European Research Area
ERC	European Research Council
ESFRI	European Strategy Forum on Research Infrastructures
ESI Europe	Energy Savings Insurance in Europe (H2020 project)
ETSI	European Telecommunications Standards Institute
EU	European Union
EUOS	EU Observatory for ICT Standardisation (H2020 project)
EV	Electric Vehicle
EVEARA	H2020 project
FENTEC	Functional Encryption Technologies project (H2020 project)
FET	Functional Encryption Technologies
FINSEC	Integrated Framework for Predictive and Collaborative Security of Financial Infrastructures (H2020 project)
FISHY	Coordinated framework for cyber resilient supply chain systems over complex ICT infrastructures (H2020 project)
FOrLedger	Blockchain-based Middleware Platform for Food Tracking Ledger Builder
FTI	Fast Track to Innovation
GDC	Genetic Data CUBE (H2020 project)
GDM	Data Marketplace deployment (H2020 project)
GDPR	General Data Protection Regulation
GOIN	H2020 project
GRECA	H2020 project
H2020	Horizon 2020
HEIF	Higher Education Innovation Fund
HEP	Higher Education Providers
HMCS	Handheld Molecular Contaminant Screener (H2020 project)
HPC	High-Performance Computing
IA	Innovation Action
ICARUS	Aviation-driven Data Value Chain for Diversified Global and Local Operations (H2020 project)
ICN	Information-Centric Networking
ICONET	New ICT infrastructure and reference architecture to support Operations in future PI Logistics NETworks (H2020 project)
I-Corps	Innovation Corps
ICT	Information and Communication Technologies
IETF	Internet Engineering Task Force
IMI2-RIA	Innovative Medicines Initiative 2 - Research and Innovation Action
INEA	(European Commission's) Innovation and Networks Executive Agency
INNOSUPSME	Innovation Support for SME
IoT	Internet of Things
IP	Intellectual Property
IRTF	Internet Research Task Force
ISO	International Organization for Standardization
ITA	Innovations and Technologies Analysis
KIOS CoE	KIOS Research and Innovation Center of Excellence
KYC	Know Your Customer
LAN	Local Area Network
LPS	Laptop Pocket Server
MDOT	Medical Device Obligations Taskforce (H2020 project)
MH-MD	My Health - My Data (H2020 project)
MSCA	Marie Skłodowska-Curie Actions
NECOS	Novel Enablers for Cloud Slicing (H2020 project)
NeTS	Networking Technology and Systems
NFV	Network Functions Virtualisation
NGI	Next Generation Internet
NGO	Non-governmental Organisation
NIMBLE	Collaboration Network for Industry, Manufacturing, Business and Logistics in Europe (H2020 project)

NIS	Network and Information Systems
NRG-5	Enabling Smart Energy as a Service via 5G Mobile Network advances (H2020 project)
NSF	National Science Foundation
PARITY	Pro-sumer AwaRe, Transactive Markets for Valorization of Distributed flexibilITY enabled by Smart Energy Contracts (H2020 project)
PCCE	Pricing carbon with a dedicated currency to empower economic agents (H2020 project)
PCHP	PeachPie Compiler Platform (H2020 project)
PCP	Pre-Commercial Procurement
PDL	Permissioned Distributed Ledger
PHP	PHP Hypertext Preprocessor
PLM	Project Lifecycle Management
PPP	Public-Private Partnership
PV	Photo-Voltaic
RCM	Rail-Care Maintenance
RESISTO	RESIlience enhancement and risk control platform for communication infraSTructure Operators (H2020 project)
RIA	Research and Innovation Action
RISE	Research Center on Interactive Media, Smart System and Emerging Technologies (H2020 project)
SAGRIS	Sentinels-based Agriculture Information Service Component (H2020 project)
SaTC	Secure and Trustworthy Cyberspace
SBIR	Small Business Innovation Research
SDN	Software Defined Network
SELIS	Towards a Shared European Logistics Intelligent Information Space (H2020 project)
SERUMS	Securing Medical Data in Smart Patient-Centric Healthcare Systems (H2020 project)
SESAR-RIA	Single European Sky ATM Research - Research and Innovation Action
SGA-CSA	Specific Grant Agreement - Coordination and Support Action
SHAR-Q	Storage capacity sharing over virtual neighbourhoods of energy ecosystems (H2020 project)
SIEM	Security Information Event Management
SKIN	Short supply chain Knowledge and Innovation Network (H2020 project)
SLA	Service-Level Agreement
SMASH	Smart Sharing (H2020 project)
SME	Small- and Medium-sized Enterprise
SME-1	Small- and Medium-sized Enterprise Instrument Phase 2
SME-2	Small- and Medium-sized Enterprise Instrument Phase 2
SME-2b	Small- and Medium-sized Enterprise Instrument (grant only and blended finance)
SMESEC	Protecting Small and Medium-sized Enterprises digital technology through an innovative cyber-SECurity framework (H2020 project)
SOFIE	Secure Open Federation for Internet Everywhere (H2020 project)
SOTER	cyberSecurity Optimization and Training for Enhanced Resilience in finance (H2020 project)
SPHINX	Universal Cyber Security Toolkit for Health-Care Industry (H2020 project)
STMS	Smart Tyre Management System for Safer, Greener and More Economic Transport (H2020 project)
STTR	Small business Technology Transfer
TALENT SWARM	H2020 project
TCBL	Textile and Clothing Business Labs Transformative Business Models for the Textile Clothing Sector (H2020 project)
TEN-T	Trans-European Transport Network
TNT	Truth-not-Trust (H2020 project)
TOKEN	Transformative Impact Of BlocKchain tEchnologies iN Public Services (H2020 project)
TOOP	The Once Only Principle
TRESCA	Trustworthy, Reliable and Engaging Scientific Communication Approaches (H2020 project)
TSO	Transmission System Operators (for electricity supply)
TWG BLOCK	Technical Working Group BLOCKchain (of StandICT.eu project)
UK	United Kingdom
US	United States
VAT	Value-Added Tax
WLAN	Wireless LAN
ZSM	Zero touch network & Service Management

4 PDL Research and Innovation Landscape

4.1 Introduction

The present document discusses the current research and innovation programmes related to permissioned distributed ledgers, distributed digital ledger technologies and blockchain with the goal of identifying advanced technologies and innovative research results relevant or essential to PDL standardization. Without limitation to other research and innovation initiatives, the present document focuses on PDL research and innovation activities funded under the European Union's Research and Innovation Framework Programme Horizon 2020.

4.2 Related Documents and Information Sources

The present document relates to the Permissioned Distributed Ledger (PDL) Landscape of Standards and Technologies Annex B: List of EU funded Horizon 2020 Research and Innovation Projects on Blockchain and/or Distributed Ledger (ETSI GR PDL 001 [i.1]).

For the identification of Horizon 2020 Research Projects related to blockchain and DLT the Community Research and Development Information Service (CORDIS) [i.2] has been used. CORDIS serves as the European Commission's primary source of information on results from projects funded under the European Research Framework Programmes. CORDIS itself is governed and funded as part of the EU Horizon 2020 research framework programme.

The CORDIS public repository with Horizon 2020 project information held by the European Commission provides search functionalities that have been used to identify projects that have blockchain or distributed ledger technologies in any of their objectives, work program, deliverables, results or publications mentioned.

5 European Research Programmes and EU Standardization

5.1 Research

5.1.1 Research introduction

In Europe, research and innovation has a high priority. The EU provides funding for research projects in Europe within the EU-Framework Programmes for Research. These Programmes have been established since 1984 with the overall objective to strengthen the competitiveness and growth of the European market.

5.1.2 Horizon 2020

Horizon 2020 is the 8th EU Framework Programme for Research and Innovation. Reflecting on the growing importance of innovation from previous EU Framework Programmes for Research the Horizon 2020 Framework Programme has a strong focus on Innovation. In particular Horizon 2020 has implemented instruments for scaling-up the ICT innovation ecosystem in Europe. Support for ICT innovation plays a central role in Horizon 2020 together with the creation of better framework conditions for innovation in Europe through ICT standardization. The ICT innovation strategy is to ensure that the rapidly changing ICT technology directly transforms into substantial benefits for European citizens and society.

Horizon 2020 is funding research projects covering the whole innovation chain from foundational research towards the preparation of market-ready products. With a total budget of about 80 billion euros over a runtime of seven years, i.e. from 2014 to 2020, Horizon 2020 has been one of the largest research and innovation funding programmes worldwide.

- Horizon 2020 is structured into the three research and innovation programme priorities: Excellent Science
- Leadership in Enabling and Industrial Technologies
- Societal Challenges

An overview of the overall EU Horizon 2020 signed grants over the period from 2014 to 2020 is provided in Figure 5.1, whereby the information was retrieved from the EU Horizon Dashboard website https://webgate.ec.europa.eu.

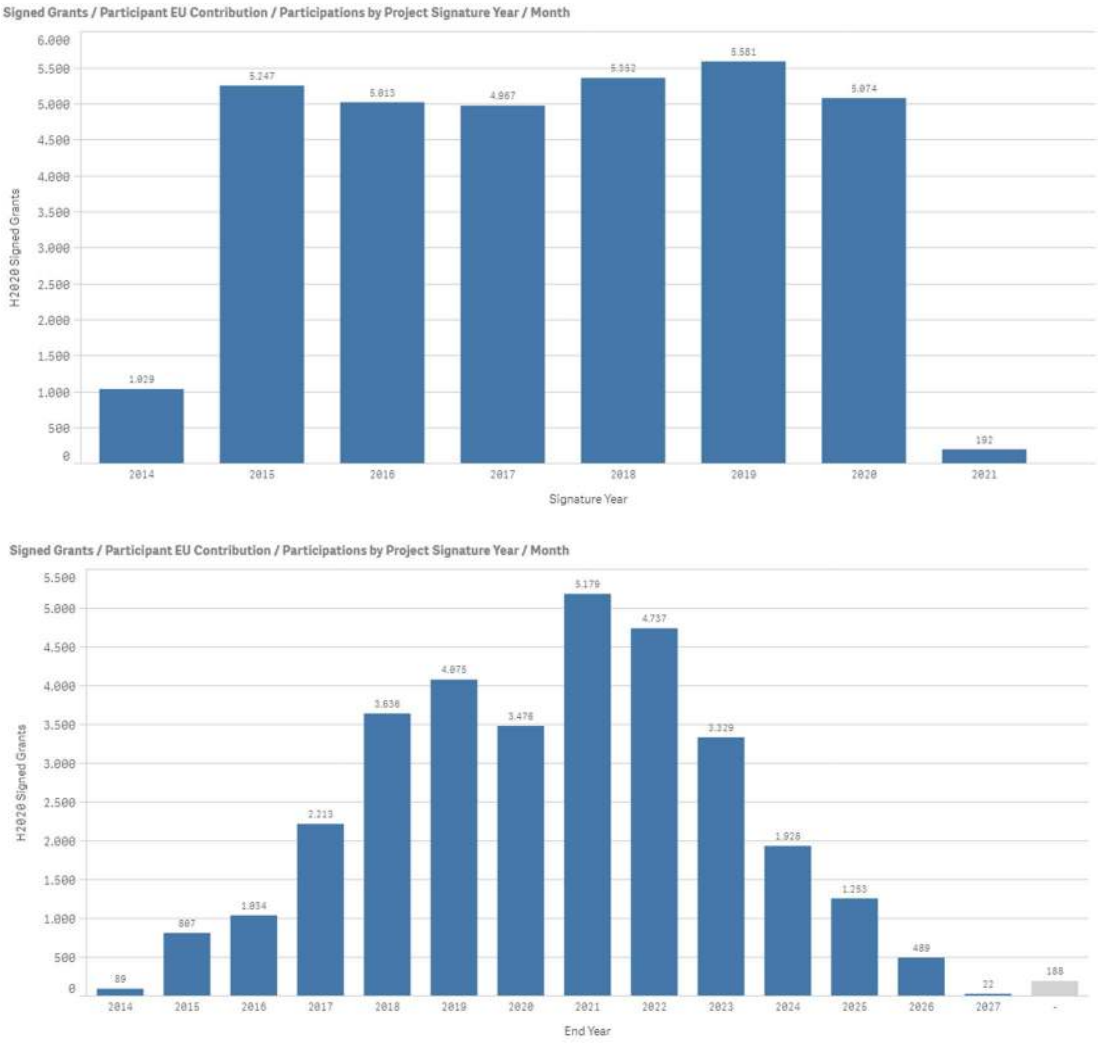


Figure 5.1: Overview on EU Horizon 2020 signed grants by signature year and end year

5.1.3 Horizon Europe

Horizon Europe is the successor of Horizon 2020 and will be the largest EU Research and Innovation Framework Programme following up on the implementation and maintenance of well-established programmes from Horizon 2020 with a budget of around 95,5 billion euro over a runtime of seven years from 2021 until 2027.

Horizon Europe is set out to strengthen research and innovation in Europe further and to drive the digital transformation for supporting the creation of innovative services and new markets and to provide more targeted solutions to global and societal challenges. Horizon Europe has the objective to create impact more effectively through a clear mission orientation and a strategy towards active engagement and involvement of society and citizens as well as stronger dissemination and exploitation of research and innovation results. With an exclusive focus on civil applications the specific programme implementing Horizon Europe is structured into the three priority areas:

- excellent science;
- global challenges and European industrial competitiveness; and
- innovative Europe.

The 'Excellent Science' work programme comprises 'European Research Council' (ERC), 'Marie Skodowska-Curie' and 'Research Infrastructures' actions set up to strengthen excellent European science and technology through more investments into individual and highly skilled researchers and leading-edge innovations.

The 'Global Challenges & European Industrial Competitiveness' work programme is set-up to advance the industrial competiveness and innovation capacities of the EU. It encompasses funding of research activities through clusters relating to global challenges (i.e. 'Health'; 'Culture, Creativity and Inclusive Society'; 'Civil Security for Society'; 'Digital Industry and Space', 'Climate, Energy and Mobility'; 'Food, Bio economy, Natural Resources, Agriculture and Environment') and also includes support activities by the 'Joint Research Centre' for policies.

The 'Innovative Europe' work programme is grouped into 'European Innovation Council' (EIC), 'European Innovation Ecosystems' and 'European Institute of Innovation and Technology' (EIT) activities. The Horizon Europe Research and Innovation Programme is based on the Widening Participation and Strengthening the European Research Area (ERA).

In order to provide solutions to major societal challenges, research and innovation missions will be an integral part of the Horizon Europe Framework Programme. Missions and related actions will be launched under Horizon Europe in the five areas:

- 'Adaptation to climate change including societal transformation'
- 'Cancer'
- 'Climate-neutral and smart cities'
- 'Healthy oceans, seas, coastal and inland waters'
- 'Soil health and food'

The missions will be implemented through a range of dedicated actions from research projects, policy measures towards legislative initiatives thus complementing the scope and effectiveness of individual actions and contributing to the Sustainable Development Goals, to the goals of the European Green Deal and the Beating Cancer Plan.

5.1.4 Other Research Funding Programmes

The Connecting Europe Facility (CEF) run by the European Commission's Innovation and Networks Executive Agency (INEA) is a fund for European infrastructure investments in transport, energy as well as digital projects leveraging not only intra- but also cross-border connectivity within and between the EU member states. CEF funding and grants are organized through calls for proposals are particularly set out to support the adoption of the CEF Building Blocks.

CEF Building Blocks are most commonly needed digital services based on European legislation and standards helping to implement the development and adoption of the required digital infrastructures in a more efficient way by using existing technologies and synergies. The European Commission's Directorate-General for Communications Networks, Content and Technology (DG CNECT) is responsible for managing the Digital Agenda and is defining the CEF Building Blocks. Among the currently available CEF Building Blocks including eID, eDelivery, eInvoicing, eSignature, Context Broker, eArchiving, eTranslation and the Big Data Test Infrastructure (BDTI), in the context of DLT related Research and Innovation programmes the CEF Building Block European Blockchain Services Infrastructure (EBSI) needs to be highlighted. The goal of the CEF EBSI also described in ETSI GR PDL 001 [i.1] is to provide cross-border pan-European public services using blockchain technology with the highest standards of security and privacy.

In November 2020, the EC launched a call for tenders on EU Blockchain Pre-Commercial Procurement (PCP) [i.3] for novel, innovative distributed ledger or blockchain solutions helping to establish the future European Blockchain Services Infrastructure for the European Citizens in compliance with the EU legal framework (i.e. GDPR, eIDAS and NIS Directive). In particular the PCP call is set out to support a number of new types of use cases with high-volume and high-velocity requirements improving interoperability, security, robustness and sustainability of EBSI. The European Blockchain PCP will award Research and Development service contracts to a number of blockchain solution providers for the development of innovative solutions covering all phases from design, prototyping towards installation and testing. Among others these solutions should provide for identification and traceability of objects, and the management of their data, automation of tasks through smart contracts, also relevant interfaces for the exchange and interoperability with external solutions e.g. IoT and AI , improving scalability and providing security levels addressing new cyber threats, addressing potential use cases like digital product passport within the circular economy, tracking of digital records and associated rights or other IoT use cases. Through the European Blockchain PCP call for tender around 7 contractors, which may include consortia, will be funded progressively in three different phases of 3, 6 and 12 months. The first phase launched in April/May 2021.

5.2 Standardization

5.2.1 European ICT Standardization

There are a number of ongoing individual as well as joint efforts and programs to bring innovative solutions and research results related to distributed ledger technologies and blockchain into standardization activities of the major and relevant standardization organizations. The following clauses highlight a few research and standardization support and coordination activities bringing together the relevant research and standardization communities.

A description of the related SDOs and their scope of work can be found in ETSI GR PDL 001 [i.1] Landscape of Standards and Technologies.

The European Union Regulation (1025/2012) [i.4], adopted by the European Parliament and by the Council of the EU, established the current legal framework for standardization. It entered into force on 1 January 2013. The three European Standardization Organizations, CEN, CENELEC and ETSI are officially recognized as the competent authorities in the area of voluntary technical standardization, including ICT.

ETSI GR PDL 001 [i.1] provides some overview of various standardization activities. Regarding all activities in ETSI, the annual report [i.4] covers past results to end of 2019 and the ETSI work programme 2020-21 gives an overview of ongoing an upcoming activities.

5.2.2 Blockchain Standardization

Within ETSI, the key group for distributed ledger technology is ISG PDL (source of the present document).

Within CEN/CENELEC, the key group is CEN/CLC/JTC 19 for Blockchain and Distributed Ledger Technologies.

There are several activities of the EC to monitor and promote distributed ledger standardization, for example:

- EU Blockchain Observatory and Forum https://www.eublockchainforum.eu.
- European Blockchain Partnership https://digital-strategy.ec.europa.eu/en/policies/blockchain-partnership.
- EC Roundtable ICT Vertical and Horizontal Projects for Blockchain Standardization.
- StandICT.eu EU Observatory for ICT Standardisation (EUOS), TWG BLOCK https://www.standict.eu/discussion-groups/twg-block-blockchain/302.

5.2.3 EC Rolling Plan 2021 for ICT Standardization

Standardization is a means, not an end. In particular, standards are an essential component of European policies for the digital transformation. The EC publishes annually, in collaboration with stakeholders and standardization bodies, a "Rolling Plan" review of standards areas which are relevant to current policies and that identifies where possible gaps where further work would be advisable [i.5]. In particular, the Rolling Plan 2021 contains a special five-page section "Blockchain and distributed ledger technologies" which concludes with a list of eight requested actions.

Particularly relevant for the ISG PDL work are Actions 5 and 6:

- ACTION 5: Standardization of the operation and reference implementation of permissioned distributed ledgers and distributed applications, with the purpose of creating an open ecosystem of industrial interoperable solutions.
- ACTION 6: Standards Development Organizations active in blockchain/DLT standardization to liaise and coordinate to take advantage of synergies and maximize resources, including with relevant public and private partnerships.

Additionally, the following points are clearly within the scope of ISG PDL:

- ACTION 1: The standardization community should continue analysing possible standardization gaps and reflect on best way to fill them. [...]

- ACTION 3: Continue identifying use cases which are relevant for EU (including EU regulatory requirements like from GDPR, ePrivacy, eIDAS, TOOP, etc.) and submit them to relevant standardization bodies, including CEN-CENELEC and ETSI, and also ISO, ITU.
- ACTION 7: A general framework for Governance of the European networks based on DLT should be developed to allow the flow of smart contracts between different networks.
- ACTION 8: ESOs to develop the standards needed for the introduction of a programmable Euro (CBDC) and token economy (upcoming MiCA Regulation), in particular to ensure interoperability with smart-contracts, legacy systems, etc.

6 PDL related Research and Innovation Activities

6.1 Introduction

In this clause an overview of public research and innovation programmes and grants related to BDLT/PDL research topics and related thematic priorities and challenges is provided. Starting from an European level with the Horizon 2020 Research Programme and the Horizon 2020 Work Programme 2018-2020, Information and Communication Technologies [i.6], is analysed with respect to research topics and programme objectives related to PDL application domains.

The focus of the following clauses is then on selected national research and innovation funding programmes and projects related to BDLT/PDL; in particular there is an overview on recent as well as upcoming BDLT related public national research programmes and investments in the US; UK, Germany, and Spain.

Readers should be aware that no overview remains current very long and this clause can only provide a partial insight.

6.2 Horizon 2020 Projects related to DLT and Blockchain

A comprehensive list of EU Horizon 2020 Projects related to DLT and blockchain can be found in the Annex A, Table A.1 of the present document. The information on the projects in this table has been retrieved from the CORDIS data base. The majority of the projects is funded under the Research and Innovation Action (RIA) schemes, followed by the Small- and Medium-sized Enterprise (SME) instrument funding schemes, then by the Innovation Action (IA) schemes and finally the Coordination and Support Action (CSA) funding schemes. This is illustrated in Figure 6.1 below, where the left figure shows the distribution of all the projects listed in annex A, Table A.1 and the right figure shows DLT related grants under the MSCA and ERC funding scheme.

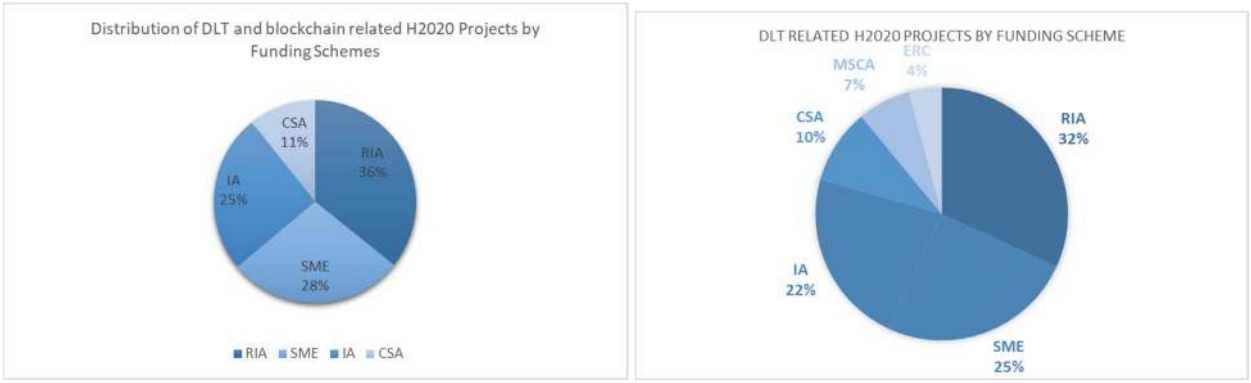


Figure 6.1: BDLT related projects by funding scheme

The PDL Application Domain or Field of Research of the Horizon 2020 projects related to blockchain and DLT is distributed as shown in the Figure 6.2, which shows the distribution of all the projects listed in Table A.1.

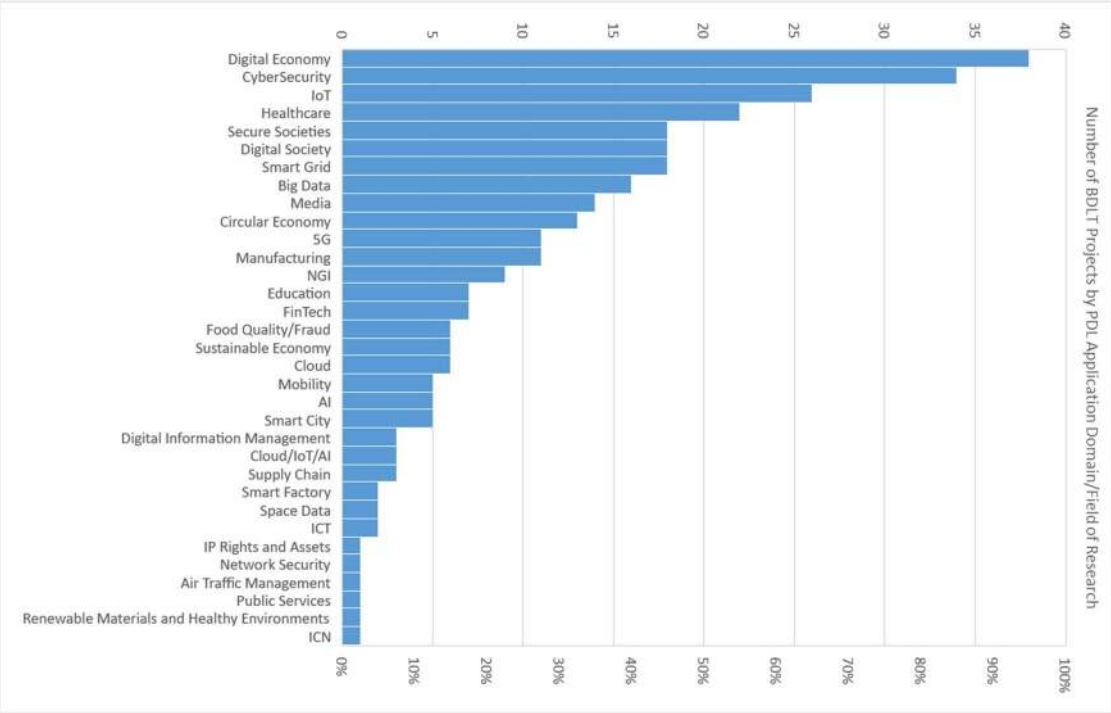


Figure 6.2: BDLT related projects by PDL Application Domain/Field of Research

6.3 Research and Innovation Programmes related to PDL

6.3.1 Scope

This clause provides some comparisons of European research programmes, EU national research programmes and some similar projects in other jurisdictions.

6.3.2 Research Programmes: EU

Within the EC European Research Council 2021 Work Programme [i.7] there are three dedicated research topics with thematic focus on Blockchain under these three blockchain research topics seven H2020 grants with a total volume of around 32,8 million euros have been provided by the EU. These seven blockchain projects relate to three thematic priorities/pillars of the H2020 Programme:

- Information and Communication Technology (ICT);
- Innovation Support for SME (INNOSUPSME); and
- Health (HEALTH).

Table 6.1: H2020 Blockchain Research Topics

Topic Code	Research Topic	Thematic Priority	H2020 Projects	H2020 EU Contribution
ICT-54-2020	Blockchain for the Next Generation Internet	ICT	3	€ 20 066 094
INNOSUP-03-2018	Blockchain and distributed ledger technologies for SMEs	INNOSUPSME	3	€ 4 493 900
IMI2-2018-15-02	Blockchain Enabled Healthcare	HEALTH	1	€ 8 290 694
Total	3/147		7	€ 32 850 688
NOTE 1: The data in this table is based on the analysis of the Research Topics of the 316 EU Horizon 2020 Projects related to DLT and Blockchain provided in Table A.1. NOTE 2: No guarantee can be given for the completeness, correctness and topicality of the information and contents compiled here				

According to CORDIS, the H2020 projects in Table A.1 relate to 147 distinct H2020 Research Topics which are listed in Table B.1. Three of the 147 H2020 Research Topics are the Blockchain Research Topics listed in Table 6.1. From the 147 Research Topics, thirty three are Research Topics of the thematic Priority ICT.

An overview of the Horizon 2020 Projects in Annex A related to research topics with thematic Priority ICT is provided in Figure 6.3.

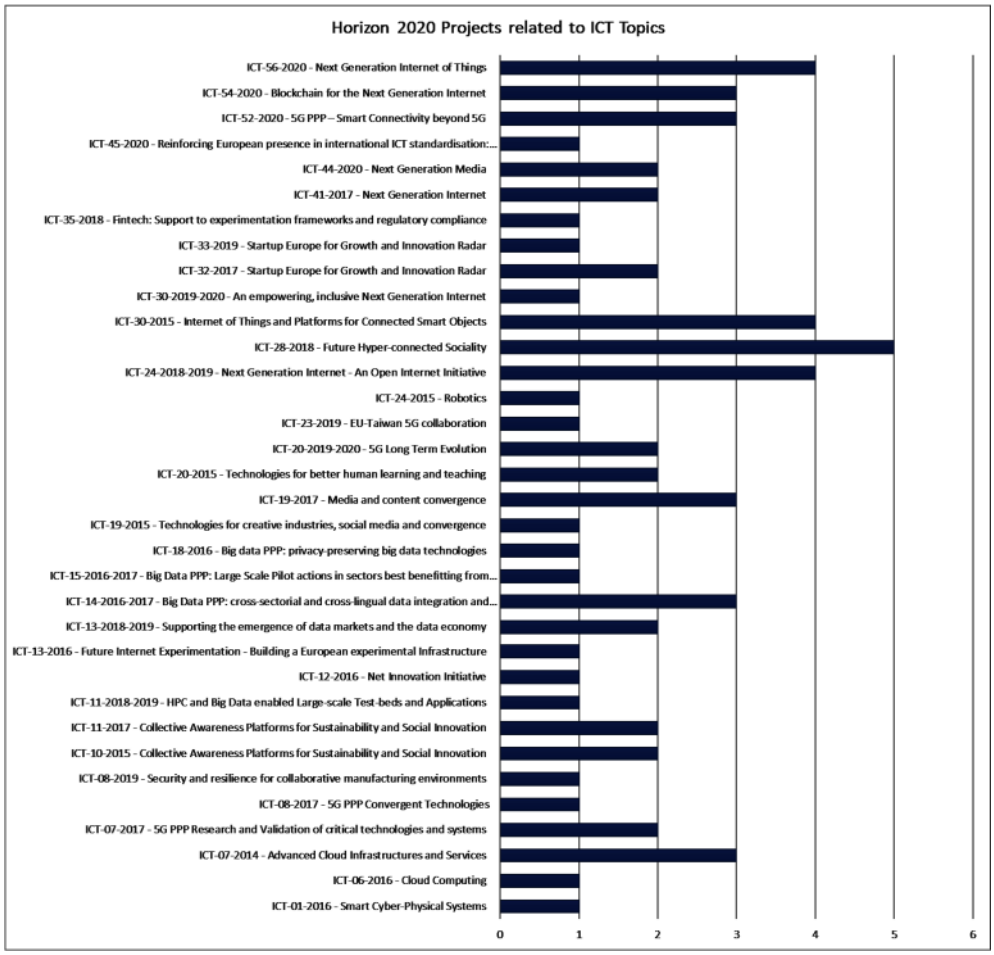


Figure 6.3: Horizon 2020 projects related to ICT Topics

An overview on BDLT H2020 Projects by thematic priorities and Research Topics: Cybersecurity, 5G; Cloud; IoT, AI, etc. is shown in Figures 6.4 and 6.5.

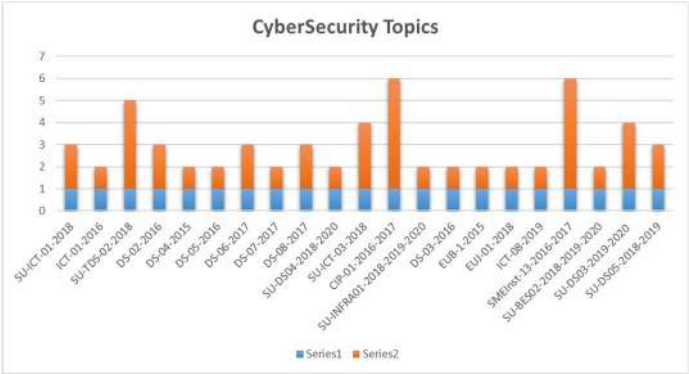


Figure 6.4: Horizon 2020 projects related to Cybersecurity Research Topics



Figure 6.5: Horizon 2020 projects related to 5G Research Topics

Finally, the very wide span of topics influenced by DLT and PDL can be seen by analysing Annex A into the form of a mindmap of broad categories as shown in Figure 6.6, which can be more finely distinguished as for example in Figure 6.7.

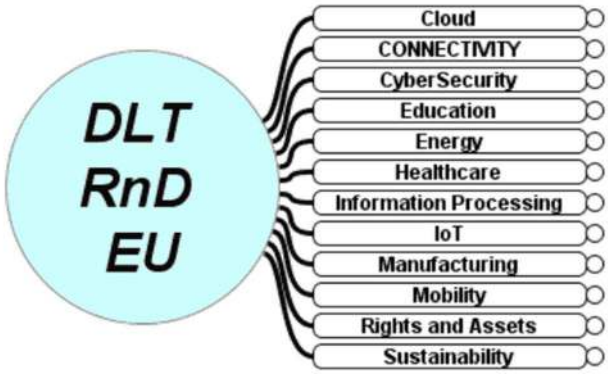


Figure 6.6: A mindmap of DLT projects in Horizon 2020, under broad categories

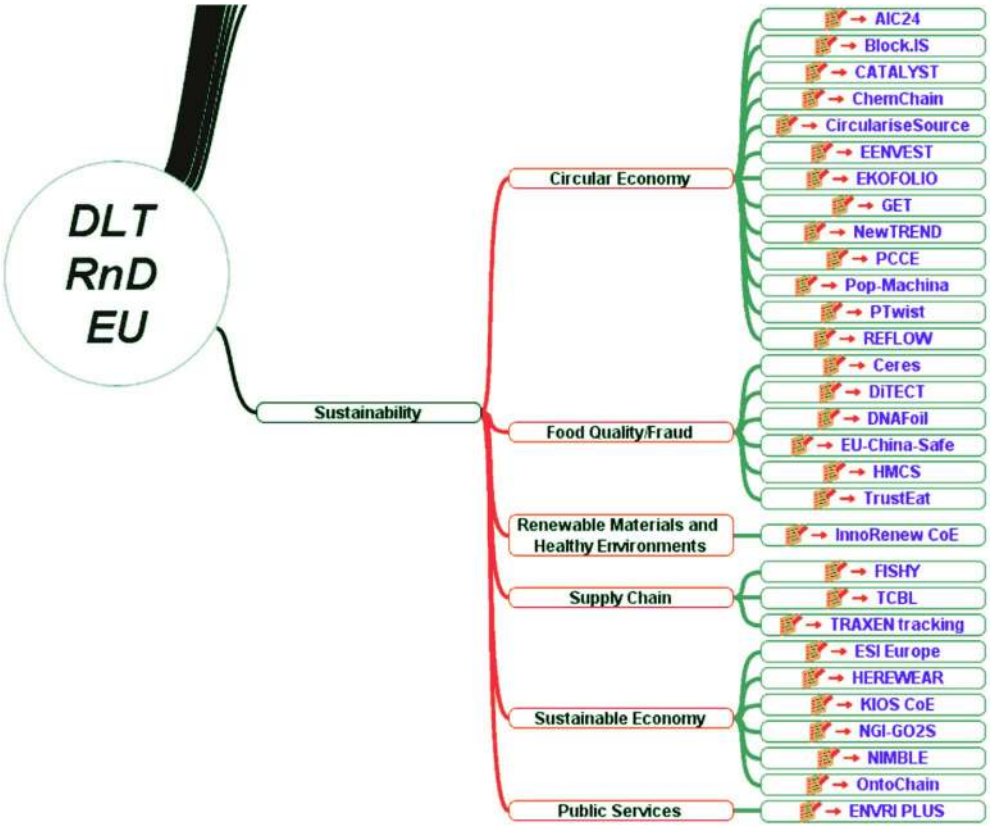


Figure 6.7: Some details of the mindmap of Figure 6.6, for Horizon 2020 projects

6.3.3 Research Programmes: Germany

In 2019 the Federal Government of Germany adopted a comprehensive "Blockchain - Strategy" [i.8] to strengthen and advance an innovative blockchain ecosystem in Germany further. The blockchain strategy of the German federal government comprises 44 measures in five priority areas of action to foster the development and scaling of blockchain applications. It entails related strategic investments in these areas to be launched by the end of 2021. With its blockchain strategy, the German federal government aims to exploit the opportunities offered by the blockchain technology and intends to activate its potentials for driving the digital transformation.

National public funding programmes for research projects related to blockchain and DLT have been established among others by the German Federal Ministry of Education and Research (BMBF - Bundesministerium für Bildung und Forschung). The BMBF is funding research on blockchain-based verification of university education certificates (Blockchain in Education).

The most recent research funding programme of the BMBF, announced in January 2021, is the research funding framework programme "Insight - Interdisciplinary perspectives of the societal and technological change" formerly also known as "ITA - Innovations - and Technologies Analysis". The Insight research framework programme is a strategic instrument and has the objective to analyse and evaluate the perspectives and challenges of new societal and technological developments and innovations in order to produce research results relevant for shaping politics and future strategies and policies. The Insight research programme has a holistic approach towards research and the funding comprises besides the scientific and technological aspects of research also ethical, social, legal, economic and political aspects in order to allow for an impact as well as potential risk assessment of innovative solutions in several dimensions. The Insight programme also has a focus on communication and direct participation and involvement of citizens. The funding is directed towards multi-perspective scientifically interdisciplinary research projects, in order to develop several alternative solution options and recommendations for action, addressing future challenges as well as opportunities of technological and social innovations. In particular, the BMBF will be funding research projects on innovation of-political activities, with thematic topics Application of Blockchain, Digital Education, Hydrogen, and impact assessment of social innovation scenarios.

6.3.4 Research Programmes: United Kingdom

UK Research Projects on application of PDL/DLT/Blockchain/Smart Contract enabling IoT adoption include the following.

The Engineering and Physical Sciences Research Council (EPSRC) (https://epsrc.ukri.org/) is a British Research Council that provides government funding for grants to undertake research and postgraduate degrees in engineering and the physical sciences, mainly to universities in the United Kingdom. EPSRC Projects include: "Accountable Just-in-Time Resource Reservation with Smart Contracts". E.g. King's College London, UK has contributed a paper under this project, on "AJIT: Accountable Just-in-Time Network Resource Allocation with Smart Contracts" (https://dl.acm.org/doi/pdf/10.1145/3411043.3412506), which was presented at the MobiArch- 15th Workshop on Mobility in the Evolving Internet Architecture (London, UK September 25, 2020) (http://cosafe.org.uk/mobiarch2020.html).

Research England's Connecting Capability Fund (CCF) (https://re.ukri.org/knowledge-exchange/the-connecting-capability-fund-ccf/) complements Higher Education Innovation Fund (HEIF) by supporting Higher Education Providers' (HEP) collaboration in commercialization through competitive projects and formula funds. The CCF aims to share good practice and capacity internally across the higher education sector, forge external technological, industrial and regional partnerships, and deliver the Government's Industrial Strategy priorities. The CCF funds among other projects the Pitch-In "Connecting Capabilities for the Internet of Things" project (https://pitch-in.ac.uk/), a research collaboration project between the Universities of Cambridge, Newcastle, Oxford and Sheffield, UK, which is connecting capabilities to enable the Internet of Things adoption.

BEAT, Blockchain-Enabled Accountable and Transparent Infrastructure Sharing in 6G: BEAT introduces an end-to-end accountable and transparent infrastructure sharing architecture e blockchains and smart contracts. BEAT enables multi-vendor and multi-operator operations by solving the challenges of a transparent sharing of infrastructure and network resources by means of PDL smart contracts. A PDL is formed among consortium members owning or wanting access to networking resources. In BEAT, Service Level Agreements (SLAs) are recorded as smart contracts, which automatically record any networking and API flow details to monitor and enforce the SLAs between the consortium members.

Pitch-in Project: https://pitch-in.ac.uk - this project give presentation to the PDL group.

AJIT: Accountable Just-in-Time Network Resource Allocation with Smart Contracts: with the surge in demand for network resources, operators need to cope with this grown demand. In future, more devices are expected to request network services and more often than before, this means operators need an automated system to accommodate these requests. Moreover, the requirement for differentiated services adds an additional challenge. Operators need to rethink current resource allocation methods and should introduce more flexible, short-term and tailor-made service offerings considering their coverage and capacity, and if and only if they can guarantee the Service Level Agreement (SLA) promised.

Providing flexible, dynamic network connectivity along with strict and guaranteed SLAs for millions, probably billions, of devices is not an easy task. This approach will cause substantial management overheads, and customers will need to contact operators more frequently for new service contracts, causing additional staff and operating expenses. Another problem is that SLAs are often opaque to the customer, making it difficult for them to detect an infringement.

AJIT exploits the inherent transparency, automation, and immutability properties of Distributed Ledger Technology (DLT) to solve the issues above. Service contracts are deployed as smart contracts on a Permissioned Distributed Ledger (PDL) that records all the service provisioning and can provide an impartial record for SLA accountability. AJIT provides an end-to-end system architecture for dynamic resource provisioning with smart contracts and side channels [i.9], [i.10] and [i.11].

BCDMA - Blockchain Division Multiple Access for IEEE 802.11: Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) has become the de facto multiple access mechanism for Wireless Local Area Network (WLAN). WLAN introduces an element of flexibility into LAN architecture, providing mobility to users, while still ensuring network access via a wireless medium. It is built on contention for the transmission medium. With contention, there is bound to be collisions, so a necessary step was the inclusion of special frames which avoid collision but significantly lower the bandwidth utilization.

To this end, and with interest by UK industry and regulator, a novel multiple access protocol titled BCDMA has been introduced [i.12], [i.13] and [i.14]. In essence, BCDMA allocates access to the transmission medium using dedicated time slots. This allocation ensures a certain level of throughput due to devices transmitting strictly within their time slot.

BCDMA also has a higher goodput, given that there are no retransmissions as a result of collisions. By treating the time-slots as assets on a smart contract, BCDMA allocates time-slots to devices.

Compared to CSMA/CA, the proposed BCDMA protocol yields improvements of more than 85 % throughput when the load capacity of the network is near saturation. Furthermore, there are qualitative improvements in terms of security with regards to authentication within the WLAN. By using PDL, BCDMA is not prone to the attack rate experienced in public networks.

Standardization is essential to adopt smart contract as SLAs, because this will enable automated, transparent and automated service contracts.

6.3.5 Research Programmes: United States of America

Table C.1 includes a list of selected ongoing blockchain-related projects in the United States (US), which have been funded by the National Science Foundation (NSF). These projects were funded through various NSF programs such as Faculty Early Career Development (CAREER), Innovation Corps (I-Corps), Networking Technology and Systems (NeTS), Secure and Trustworthy Cyberspace (SaTC), Small Business Innovation Research (SBIR), and Small Business Technology Transfer (STTR). Please note that Table C.1. does not aim to provide an exhaustive list of all blockchain-related projects in the US. But Table C.1 shows:

1) These projects cover broad technical areas from core technologies (e.g. interledger communications, payment channel networks, scalability, smart contract) to vertical domains (e.g. cybersecurity, digital economy, digital information management, fintech, healthcare, IoT, media).

2) The projects related to blockchain core technologies are funded before the projects about blockchain-enabled vertical domains.

6.3.6 Research Programmes: Spain

In Spain, the Government, regulators and prominent entities in the business field have been active to create an ecosystem since around 2017, while avoiding or warning against certain risks on several occasions e.g. initial coin offerings and digital currencies. The Spanish National Strategy for Science, Technology and Innovation (EECTI) for the 2021-2027 defines a set of strategic lines, grouped in six wider areas, as the framework for the actions of the Spanish Government to support research and technology development. Two of these areas, titled "Security for Society" and "Digitalization", include mentions to DLT and its applications as matters for research, specifically in three strategic lines: "Cybersecurity", "Artificial Intelligence and Robotics" and "Next-generation Internet".

The Council of University Rectors (CRUE) started in 2019 project BLUE, in collaboration with the public entity for the promotion of the Information Society (Red.es) and the academic network (RedIRIS). The project is focused on the applications of DLT within the academic sector, with special emphasis on academic certifications and lifelong learning support. The project started by producing a survey on the applicability of these technologies in the university environment and continued with several development projects that are consolidating into a first field trial involving three universities, integrated within the EBSI Early Adopters program, and planning cross-border pilots before the end of 2021.

The blockchain observatory provided by BI (https://blockchainintelligence.es/en/map/) lists 13 active projects working on DLT, addressing matters related with fintech, energy, legal and compliance, logistics and transportation, documents and traceability, and health. Alastria, the non-profit association promoting the development of decentralized ledger technologies and participant in ISG PDL, hosts a "project counter" (https://alastria.io/en/mesa-de-proyectos/) to support research and development initiatives in these areas.

7 Importance of PDL Standardization

Horizon 2020, the 8th EU Framework Programme for Research and Innovation, injected about 13 billion euro per year from 2015 to 2020 into the EU research ecosystem, whereas the follow on programme Horizon Europe increases that by about 20 %, with a much stronger dissemination and exploitation emphasis. The years 2015 to 2020 saw the launch of several hundred research projects involving DLT, in the areas of Digital Economy (38), Cyber Security (34), IoT (26), Healthcare (22) and so on as listed in Annex A. Various national programmes in the UK, Germany, Spain, etc, boosted the work even more.

Many of these research and innovation projects considered how to disseminate results in the marketplace, and in particular how to promote the technology by helping to create appropriate (international) standards. Two projects can serve as examples:

- SOFIE (now completed - https://www.sofie-iot.eu/news/setting-standards-for-the-future); and
- 5GZORRO [i.17].

SOFIE (Secure Open Federation for Internet Everywhere) was funded under EC contract Nr. 779984 and concluded that:

"Standardization is a framework of agreements for all relevant parties in an industry to ensure the creation of well-performing systems, products and services in accordance with set guidelines. The objective is to maximize compatibility, interoperability, safety, repeatability, and quality. [...] When it comes to blockchains and Distributed Ledger Technology (DLT), standardization is happening in all relevant technology standardization groups. Questions that we initially did not find good answers to in the SOFIE project were: What is the right way for us to contribute to standardization in a situation where the focus and maturity in blockchain adoption varied a lot between different standardization groups? Should we aim for more widespread collaboration or target our efforts to specific standardization groups? In retrospective [...]with ETSI Industry Special Group for Private Distributed Ledgers (ETSI ISG PDL) [...] active collaboration was initiated [...] by Ericsson as group member and strongly supported by Aalto University and Athens University of Economics and Business. [This work has been] complemented by contributions by other SOFIE partners also in other standardization groups, for example W3C and IETF/IRTF" [i.16].

The project 5GZORRO (Zero-touch security and trust for ubiquitous computing and connectivity in 5G networks), funded under Horizon 2020 contract 871533, started in November 2019 and has decided to "ensure the long-term success of the project through standardization and dissemination in scientific, industrial, and commercial fora, and by contributing to relevant open source communities & SDOs" [i.16]. It has as one of its core objectives:

"contributions to target SDOs/SSOs (Standards Developing Organization / Standards Setting Organization) [in ETSI ISG PDL, ETSI ISG ENI, ETSI ISG ZSM, ETSI ISG NFV, ECSO, IRTF, IETF, etc. [...] ranging from involvement in technical discussions and sharing 5GZORRO results, to active contribution to standardisation efforts and/or open-source projects and communities" [i.16].

In particular, 5GZORRO has identified the following potential collaborations with ISG PDL: PoCs on the applicability of Smart Contracts in network services, Trust computations for smart contracts and federated data, Distributed Ledger interoperability and Trusted Execution Environments, particularly in use cases such as NFV resource brokering, Spectrum trading, Service deployment and activation.

In summary for this clause and indeed for the present document, the EC has heavily promoted research in Distributed Ledger Technologies and the projects have individually seen the importance of contributing to the (international) standardization ecosystems, in order to:

- promote interoperability of solutions and "future-proof" solutions in the market place;
- improve the reliability of services;
- improve transparency of operations for service providers and ultimately for European citizens;
- offer verifiable proof or benchmarking of features such as privacy or tamper-proofing;
- ensure fair procurement by means of transparent reference to best-of-breed technological solutions (according to World Trade Office recommendations);
- improve the results of "due diligence" investigations for government and private services;

- build efficient ecosystems (and avoid vendor "lock in").

Annex A:
List of EU Horizon 2020 Projects related to DLT and Blockchain

Information about EU funded research and innovation projects under the Horizon 2020 research framework programme that are related to blockchain and/or distributed ledger technologies as obtained from CORDIS project database has been collected by ISG PDL and a list of all Horizon 2020 projects on DLT and blockchain under the funding schemes RIA, IA, CSA, as well as SME has been extracted. This list of Horizon 2020 projects related to DLT and blockchain contains for each project the following information: project acronym, project name, project ID, project funding scheme, a link to the projects factsheet in CORDIS and an indication of the projects potential application domain or field of research as specified by ISG PDL. The project list is sorted in alphabetical order by the project acronym as presented in Table A.1.

NOTE 1: The latest status update of the project information from CORDIS database is March 2021.

NOTE 2: The classification of the projects Application Domain and Field of Research is based on an ISG PDL assessment and therefore might differ from the actual projects application domain or field of research.

NOTE 3: No guarantee can be given for the completeness, correctness and topicality of the information and contents compiled here.

Table A.1: EU Horizon 2020 Projects related to DLT and Blockchain as of March 2021

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
5GaaS	5G-as-a-Service	958832	IA	https://cordis.europa.eu/project/id/958832	5G
5G-CORAL	5G-CORAL: A 5G Convergent Virtualized Radio Access Network Living at the Edge	761586	RIA	https://cordis.europa.eu/project/id/761586	5G
5G-DIVE	eDge Intelligence for Vertical Experimentation	859881	RIA	https://cordis.europa.eu/project/id/859881	5G
5G-TRANSFORMER	5G-TRANSFORMER: 5G Mobile Transport Platform for Verticals	761536	RIA	https://cordis.europa.eu/project/id/761536	5G
5GZORRO	Zero-touch security and trust for ubiquitous computing and connectivity in 5G networks	871533	RIA	https://cordis.europa.eu/project/id/871533	AI
Access2Europe	Help startups to scale by leveraging the access to European markets offered by four major startup hubs	780237	IA	https://cordis.europa.eu/project/id/780237	Digital Economy
AEGIS	Advanced Big Data Value Chain for Public Safety and Personal Security	732189	IA	https://cordis.europa.eu/project/id/732189	Big Data
AEOLIX	Architecture for EurOpean Logistics Information eXchange	690797	RIA	https://cordis.europa.eu/project/id/690797	Digital Economy
AGILE	Adoptive Gateways for dIverse muLtiple Environments	688088	RIA	https://cordis.europa.eu/project/id/688088	IoT
AI4DI	Artificial Intelligence for Digitizing Industry	826060	ECSEL-RIA	https://cordis.europa.eu/project/id/826060	AI
AIC24	Sustainable Datacenters Through Immersed Computing	816831	SME-1	https://cordis.europa.eu/project/id/816831	Circular Economy

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
AICHAIN	A platform for privacy-preserving Federated Machine Learning using Blockchain to enable Operational Improvements in ATM	894162	SESAR-RIA	https://cordis.europa.eu/project/id/894162	Digital Information Management
AMable	AdditiveManufacturABLE	768775	IA	https://cordis.europa.eu/project/id/768775	Manufacturing
AMBRPAY	Decentralized subscription payments with cryptocurrency	887968	SME-1	https://cordis.europa.eu/project/id/887968	Digital Economy
ANITA	Advanced tools for fighting oNline llegal TrAfficking	787061	RIA	https://cordis.europa.eu/project/id/787061	CyberSecurity
ARCAone	Next generation security platform to safeguard critical applications and sensitive digital assets	888406	SME-1	https://cordis.europa.eu/project/id/888406	CyberSecurity
ARTICONF	smART socIal media eCOsytstem in a blockchaiN Federated environment	825134	RIA	https://cordis.europa.eu/project/id/825134	Media
ASSIST-IoT	Architecture for Scalable, Self-*, human-centric, Intelligent, Secure, and Tactile next generation IoT	957258	RIA	https://cordis.europa.eu/project/id/957258	IoT
ASSURED	Future Proofing of ICT Trust Chains: Sustainable Operational Assurance and Verification Remote Guards for Systems-of-Systems Security and Privacy	952697	RIA	https://cordis.europa.eu/project/id/952697	Network Security
ATARCA	Accounting Technologies for Anti-Rival Coordination and Allocation	964678	RIA	https://cordis.europa.eu/project/id/964678	Digital Economy
AudioCommons	Audio Commons: An Ecosystem for Creative Reuse of Audio Content	688382	RIA	https://cordis.europa.eu/project/id/688382	Media
AVANGARD	Advanced manufacturing solutions tightly aligned with business needs	869986	IA	https://cordis.europa.eu/project/id/869986	Manufacturing
B4CM	Blockchains as a Distributed Ledger for Attribution of RCM Data in Rail	826156	CSA	https://cordis.europa.eu/project/id/826156	Mobility
B4TDM	Making Contracts Digital with Civilized Blockchain	858630	SME-2	https://cordis.europa.eu/project/id/858630	Digital Economy
BD4NRG	BD4NRG: Big Data for Next Generation Energy	872613	IA	https://cordis.europa.eu/project/id/872613	Big Data
BD4OPEM	Big Data for OPen innovation Energy Marketplace	872525	IA	https://cordis.europa.eu/project/id/872525	Smart Grid
BEACON	Boosting Agricultural Insurance based on Earth Observation data	821964	IA	https://cordis.europa.eu/project/id/821964	Space Data
B-HUB FOR EUROPE	Blockchain HUB FOR EUROPEan startups acceleration and growth	871869	IA	https://cordis.europa.eu/project/id/871869	Digital Economy
BIG	Enhancing the research and innovation potential of Tecnico through Blockchain technologies and design Innovation for social Good	952226	CSA	https://cordis.europa.eu/project/id/952226	Digital Society

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
BILLON	Disrupting the economy - FinTech blockchain solution revolutionises direct payments. Secure, low-cost and simple bank-free payments for everyone	783861	SME-2	https://cordis.europa.eu/project/id/783861	FinTech
bIoTope	Building an IoT OPen innovation Ecosystem for connected smart objects	688203	RIA	https://cordis.europa.eu/project/id/688203	IoT
BitBox	Enterprise - Enterprise-grade Solution for Digital Assets Custody	855049	SME-1	https://cordis.europa.eu/project/id/855049	CyberSecurity
Bitwala	Next generation banking tools for the blockchain economy	854346	SME-1	https://cordis.europa.eu/project/id/854346	FinTech
Bitztribution	Towards a fully decentralized distribution for next generation of travel and tourism distribution	876748	SME-1	https://cordis.europa.eu/project/id/876748	Digital Economy
Block.IS	Blockchain Innovation Spaces	824509	IA	https://cordis.europa.eu/project/id/824509	Circular Economy
BlockchainKYC	Blockchain-based, 100% automated KYC (Know Your Customer) service	774802	SME-1	https://cordis.europa.eu/project/id/774802	Secure Societies
BlockchainKYC	Blockchain-based, 100% automated KYC (Know Your Customer) service	850059	SME-2	https://cordis.europa.eu/project/id/850059	Secure Societies
BLOCKCHERS	BLOCKCHAIN TECHNOLOGIES FOR SMEs	828840	CSA	https://cordis.europa.eu/project/id/828840	Big Data
Blocknetwork	Fusing Big Data and Implementing Novel Cyber Security Solutions	827457	SME-1	https://cordis.europa.eu/project/id/827457	Big Data
BLOCKPOOL	Pooling SME adoption and deployment of Blockchain and other DLTs	828888	CSA	https://cordis.europa.eu/project/id/828888	Education
BlockStart	Helping SMEs take the first steps into the Blockchain	828853	CSA	https://cordis.europa.eu/project/id/828853	Education
BLOOMEN	Blockchains in the new era of participatory media experience	762091	IA	https://cordis.europa.eu/project/id/762091	Media
BodyPass	API-ecosystem for cross-sectorial exchange of 3D personal data	779780	IA	https://cordis.europa.eu/project/id/779780	Healthcare
Bonseyes	Platform for Open Development of Systems of Artificial Intelligence	732204	RIA	https://cordis.europa.eu/project/id/732204	IoT
BOOST 4.0	Big Data Value Spaces for COmpetitiveness of European COnnected Smart FacTories 4.0	780732	IA	https://cordis.europa.eu/project/id/780732	Big Data
BRAIN-IoT	model-Based fRamework for dependable sensing and Actuation in INtelligent decentralized IoT systems	780089	RIA	https://cordis.europa.eu/project/id/780089	IoT
BrainPatch	BrainPatch - Breakthrough noninvasive brain stimulation using AI	867386	SME-1	https://cordis.europa.eu/project/id/867386	Healthcare
BRIGHT	Boosting DR through increased communIty-level consumer engaGement by combining Data-driven and blockcHain technology Tools with social science approaches and multi-value service design	957816	IA	https://cordis.europa.eu/project/id/957816	Smart Grid

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C4IIoT	Cyber security 4.0: protecting the Industrial Internet Of Things	833828	IA	https://cordis.europa.eu/project/id/833828	CyberSecurity
CATALYST	Converting DCs in Energy Flexibility Ecosystems (CATALYST)	768739	IA	https://cordis.europa.eu/project/id/768739	Circular Economy
CATTLECHAIN 4.0	Enhancing farm productivity and guaranteeing CATTLE traceability and welfare with blockCHAIN	853864	IA	https://cordis.europa.eu/project/id/853864	IoT
CEITER	Cross-Border Educational Innovation thru Technology-Enhanced Research	669074	CSA	https://cordis.europa.eu/project/id/669074	Education
Ceres	Proactive system to forecast and reduce environmental footprint while maximizing agriculture performance	832681	SME-1	https://cordis.europa.eu/project/id/832681	Food Quality/Fraud
CHARIOT	Cognitive Heterogeneous Architecture for Industrial IoT	780075	RIA	https://cordis.europa.eu/project/id/780075	IoT
ChemChain	Blockchain Platform to Track Chemicals along the Value Chain	875783	SME-1	https://cordis.europa.eu/project/id/875783	Circular Economy
CirculariseSource	Certifying recycled material while keeping market prices and allowing mass adoption	961989	SME-2b	https://cordis.europa.eu/project/id/961989	Circular Economy
CITADEL	Empowering Citizens to TrAnsform European PubLic Administrations	726755	RIA	https://cordis.europa.eu/project/id/726755	Digital Society
CITIES2030	Co-creating resIlient and susTaInable food systEms towardS FOOD2030	101000640	IA	https://cordis.europa.eu/project/id/101000640	Smart City
ClearFarm	Co-designed Welfare Monitoring Platform for Pig and Dairy Cattle	862919	IA	https://cordis.europa.eu/project/id/862919	Digital Economy
CLINICOVERY	CLINICOVERY, a versatile, high quality, environmental-friendly, easy to use e-Clinical solution for clinical research	808543	SME-1	https://cordis.europa.eu/project/id/808543	Healthcare
CO3	Digital Disruptive Technologies to Cocreate, Co-produce and Co-manage Open Public Services along with Citizens	822615	RIA	https://cordis.europa.eu/project/id/822615	Digital Society
COBAFRA	Combatting Banking Fraud with SiSid: A unique solution for preventing corporate payments fraud using AI and blockchain	835813	SME-1	https://cordis.europa.eu/project/id/835813	Digital Economy
COG-LO	COGnitive Logistics Operations through secure, dynamic and ad-hoc collaborative networks	769141	RIA	https://cordis.europa.eu/project/id/769141	IoT
COLLABS	A COmprehensive cyber-intelligence framework for resilient coLLABorative manufacturing Systems	871518	RIA	https://cordis.europa.eu/project/id/871518	Manufacturing
COMPOSITION	Ecosystem for Collaborative Manufacturing Processes - Intra- and Interfactory Integration and Automation	723145	RIA	https://cordis.europa.eu/project/id/723145	Manufacturing

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Community Cloud	Cubbit - the first distributed data-center recycling the internet resources we waste into the most competitive cloud provider.	874313	SME-2	https://cordis.europa.eu/project/id/874313	Cloud
CONCORDIA	Cyber security cOmpeteNce fOr Research anD Innovation	830927	RIA	https://cordis.europa.eu/project/id/830927	CyberSecurity
CONNECARE	Personalized Connected Care for Complex Chronic Patients	689802	RIA	https://cordis.europa.eu/project/id/689802	Healthcare
Consentio	Boosting conversations to disrupt horticulture trading	836175	SME-1	https://cordis.europa.eu/project/id/836175	Digital Economy
COOL-SENS	Advanced monitoring solution to prevent losses and assure full transparency along the cold chain	855554	SME-1	https://cordis.europa.eu/project/id/855554	IoT
CoordiNet	Large scale campaigns to demonstrate how TSO-DSO shall act in a coordinated manner to procure grid services in the most reliable and efficient way	824414	IA	https://cordis.europa.eu/project/id/824414	Smart Grid
COPA EUROPE	COllaborative Platform for trAnsmedia storytelling and cross channel distribution of EUROPEan sport events	957059	IA	https://cordis.europa.eu/project/id/957059	Media
CPN	Content Personalization Network	761488	IA	https://cordis.europa.eu/project/id/761488	Media
CREST	Fighting Crime and TerroRism with an IoT-enabled Autonomous Platform based on an Ecosystem of Advanced IntelligEnce, Operations, and InveStigation Technologies	833464	RIA	https://cordis.europa.eu/project/id/833464	Secure Societies
CRITICAL-CHAINS	IOT- & Blockchain-Enabled Security Framework for New Generation Critical Cyber-Physical Systems In Finance Sector	833326	IA	https://cordis.europa.eu/project/id/833326	CyberSecurity
CUREX	seCUre and pRivate hEalth data eXchange	826404	RIA	https://cordis.europa.eu/project/id/826404	Healthcare
CyberKit4SME	Democratizing a Cyber Security Toolkit for SMEs and MEs	883188	IA	https://cordis.europa.eu/project/id/883188	CyberSecurity
CyberSec4Europe	Cyber Security Network of Competence Centres for Europe	830929	RIA	https://cordis.europa.eu/project/id/830929	CyberSecurity
CYBER-TRUST	Advanced Cyber-Threat Intelligence, Detection, and Mitigation Platform for a Trusted Internet of Things	786698	RIA	https://cordis.europa.eu/project/id/786698	CyberSecurity
cyberwatching,eu	The European watch on cybersecurity privacy	740129	CSA	https://cordis.europa.eu/project/id/740129	CyberSecurity
D4FLY	Detecting Document frauD and iDentity on the fly	833704	RIA	https://cordis.europa.eu/project/id/833704	Secure Societies
DataVaults	Persistent Personal Data Vaults Empowering a Secure and Privacy Preserving Data Storage, Analysis, Sharing and Monetisation Platform	871755	IA	https://cordis.europa.eu/project/id/871755	Big Data

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Dawex GDM	Dawex Data Marketplace deployment: unlocking European companies' data value	829729	SME-2	https://cordis.europa.eu/project/id/829729	Digital Economy
DE4A	Digital Europe for All	870635	RIA	https://cordis.europa.eu/project/id/870635	Digital Society
DECENTER	Decentralised technologies for orchestrated cloud-to-edge intelligence	815141	RIA	https://cordis.europa.eu/project/id/815141	Cloud/IoT/AI
DECODE	Decentralised Citizens Owned Data Ecosystem	732546	RIA	https://cordis.europa.eu/project/id/732546	IoT
DEDICAT 6G	Dynamic coverage Extension and Distributed Intelligence for human Centric applications with assured security, privacy and trust: from 5G to 6G	101016499	RIA	https://cordis.europa.eu/project/id/101016499	5G
DEFENDER	Defending the European Energy Infrastructures	740898	IA	https://cordis.europa.eu/project/id/740898	CyberSecurity
DEL4ALL	Digital Enhanced Learning for ALL	871573	CSA	https://cordis.europa.eu/project/id/871573	NGI
DELTA	Future tamper-proof Demand rEsponse framework through seLfconfigured, self-opTimized and collAborative virtual distributed energy nodes	773960	RIA	https://cordis.europa.eu/project/id/773960	Smart Grid
DFS	Democratized Financial Services	865430	SME-1	https://cordis.europa.eu/project/id/865430	FinTech
DiFacturo	The Unique International Independent Decentralized Invoice Network	889930	SME-1	https://cordis.europa.eu/project/id/889930	Digital Economy
DigiJourney	Digital transformation journey for SMEs	870625	CSA-LS	https://cordis.europa.eu/project/id/870625	Big Data
DigiPharm	Value-based Healthcare Platform Secured with Blockchain Technology	888950	SME-1	https://cordis.europa.eu/project/id/888950	Healthcare
DiTECT	DIgital TEChnologies as an enabler for a conTinuous transformation of food safety system	861915	RIA	https://cordis.europa.eu/project/id/861915	Food Quality/Fraud
DLInnociate	DLInnociate Innovation Associate knowledgeable in blockchain technologyfor real time economy platform business development	739782	CSA	https://cordis.europa.eu/project/id/739782	Digital Economy
DNAFoil	Rapid DNA screening to secure the global food supply chains	836046	SME-1	https://cordis.europa.eu/project/id/836046	Food Quality/Fraud
DOMINOES	Smart Distribution Grid: a Market Driven Approach for the Next Generation of Advanced Operation Models and Services	771066	RIA	https://cordis.europa.eu/project/id/771066	Smart Grid
DrBox	eHealth platform to assess patient clinical progress due to prescribed medication	807509	SME-1	https://cordis.europa.eu/project/id/807509	Healthcare
DRIVE	Demand Response Integration tEchnologies: unlocking the demand response potential in the distribution grid	774431	RIA	https://cordis.europa.eu/project/id/774431	Smart Grid

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DSISCALE	Supporting the scale and growth of Digital Social Innovation in Europe through coordination of Europe's DSI and CAPS Networks	780473	CSA	https://cordis.europa.eu/project/id/780473	Digital Society
EBSIS	Event Based Systems in Iaşi - A Twinning between Universitatea Alexandru Ioan Cuza din Iaşi, Université de Neuchâtel and Technische Universität Dresden	692178	CSA	https://cordis.europa.eu/project/id/692178	Big Data
ECHO	European network of Cybersecurity centres and competence Hub for innovation and Operations	830943	RIA	https://cordis.europa.eu/project/id/830943	CyberSecurity
eDREAM	enabling new Demand REsponse Advanced, Market oriented and Secure technologies, solutions and business models	774478	RIA	https://cordis.europa.eu/project/id/774478	Smart Grid
EENVEST	Risk reduction for Building Energy Efficiency investments	833112	CSA	https://cordis.europa.eu/project/id/833112	Circular Economy
eFactory	European Connected Factory Platform for Agile Manufacturing	825075	IA	https://cordis.europa.eu/project/id/825075	Smart Factory
EKOFOLIO	Expanding Key Opportunities in FOrest Investments: Liquidity, Impact and Ownership through Blockchain	876676	SME-1	https://cordis.europa.eu/project/id/876676	Circular Economy
ENHANCE	Building an Excellency Network for Heightening Agricultural ecoNomic researCh and Education in Romania	691681	CSA	https://cordis.europa.eu/project/id/691681	Education
ENIT Agent 2.0	The world's first edge computing solution for SMEs enabling energy efficiency, Industry 4.0 and new business models for the energy sector	811640	SME-2	https://cordis.europa.eu/project/id/811640	Smart Grid
ENSURESEC	End-to-end Security of the Digital Single Market's E-commerce and Delivery Service Ecosystem	883242	IA	https://cordis.europa.eu/project/id/883242	CyberSecurity
ENVRI PLUS	Environmental Research Infrastructures Providing Shared Solutions for Science and Society	654182	RIA	https://cordis.europa.eu/project/id/654182	Public Services
ePIcenter	Enhanced Physical InternetCompatible Earth-frieNdly freight Transportation answER	861584	RIA	https://cordis.europa.eu/project/id/861584	Cloud/IoT/AI
ERIGrid	European Research Infrastructure supporting Smart Grid Systems Technology Development, Validation and Roll Out	654113	RIA	https://cordis.europa.eu/project/id/654113	Smart Grid
euCANSHare	An EU-Canada joint infrastructure for next-generation multi-Study Heart research	825903	RIA	https://cordis.europa.eu/project/id/825903	Healthcare
ESCUDO-CLOUD	Enforceable Security in the Cloud to Uphold Data Ownership	644579	RIA	https://cordis.europa.eu/project/id/644579	Cloud

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ESI Europe	Driving Investment in Energy Efficiency through Energy Savings Insurance in Europe	785061	CSA	https://cordis.europa.eu/project/id/785061	Sustainable Economy
EU-China-Safe	Delivering an Effective, Resilient and Sustainable EU-China Food Safety Partnership	727864	RIA	https://cordis.europa.eu/project/id/727864	Food Quality/Fraud
EUNOMIA	User-oriented, secure, trustful & decentralised social media	825171	IA	https://cordis.europa.eu/project/id/825171	Media
European IPR Helpdesk	The European IPR Helpdesk	641474	CSA	https://cordis.europa.eu/project/id/641474	IP Rights and Assets
EVEARA	EVEARA: Testing a New Business Model for Digital Music Distribution	735359	SME-1	https://cordis.europa.eu/project/id/735359	Digital Economy
EyeCode	A frictionless two-factor authentication software for secure transactions	816804	SME-1	https://cordis.europa.eu/project/id/816804	CyberSecurity
FAR-EDGE	Factory Automation Edge Computing Operating System Reference Implementation	723094	RIA	https://cordis.europa.eu/project/id/723094	Manufacturing
FASTER	First responder Advanced technologies for Safe and efficienT Emergency Response	833507	RIA	https://cordis.europa.eu/project/id/833507	Healthcare
FeatureCloud	Privacy preserving federated machine learning and blockchaining for reduced cyber risks in a world of distributed healthcare	826078	RIA	https://cordis.europa.eu/project/id/826078	Healthcare
FENTEC	Functional Encryption Technologies	780108	RIA	https://cordis.europa.eu/project/id/780108	CyberSecurity
FEVER	Flexible Energy Production, Demand and Storage-based Virtual Power Plants for Electricity Markets and Resilient DSO Operation	864537	IA	https://cordis.europa.eu/project/id/864537	Smart Grid
FINSEC	Integrated Framework for Predictive and Collaborative Security of Financial Infrastructures	786727	IA	https://cordis.europa.eu/project/id/786727	CyberSecurity
FIN-TECH	A FINancial supervision and TECHnology compliance training programme	825215	CSA	https://cordis.europa.eu/project/id/825215	FinTech
FISHY	A coordinated framework for cyber resilient supply chain systems over complex ICT infrastructures	952644	RIA	https://cordis.europa.eu/project/id/952644	Supply Chain
FLEXI-GRID	Enabling flexibility for future distribution grid	864048	IA	https://cordis.europa.eu/project/id/864048	Smart Grid
FleXunity	Scaling-up Power Flexible Communities business models empowered by Blockchain and AI	870146	IA	https://cordis.europa.eu/project/id/870146	Digital Economy
FOrLedger	A Blockchain-based Middleware Platform for Food Tracking Ledger Builder	856065	SME-1	https://cordis.europa.eu/project/id/856065	Digital Economy
GASTEJO	Decentralized Travel Apartment Distribution Platform	832537	SME-1	https://cordis.europa.eu/project/id/832537	Digital Economy

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GATEKEEPER	Smart living homes - whole interventions demonstrator for people at health and social risks	857223	IA	https://cordis.europa.eu/project/id/857223	Healthcare
GDC	A Genetic Data CUBE - An innovative business model applied to predictive and prescriptive analytics, exploring Big Data an empowering cloudservices and urgent computation	697581	SME-1	https://cordis.europa.eu/project/id/697581	Big Data
GET	Green Energy Tracker. The international tracking system for guarantees of origin, green energy certificates and green labels	775225	SME-1	https://cordis.europa.eu/project/id/775225	Circular Economy
GHOST	Safe-Guarding Home IoT Environments with Personalized Realtime Risk Control	740923	IA	https://cordis.europa.eu/project/id/740923	CyberSecurity
GLASS	SinGLe Sign-on eGovernAnce paradigm based on a distributed file exchange network for Security, transparency, cost effectiveness and truSt	959879	RIA	https://cordis.europa.eu/project/id/959879	Big Data
GOIN	Development of a fintech saving and investment application based on datadriven learning algorithms and blockchain technology	889276	SME-1	https://cordis.europa.eu/project/id/889276	FinTech
GRECA	GRECA: Revolution in corporate transactions	888213	SME-1	https://cordis.europa.eu/project/id/888213	Digital Economy
HEREWEAR	Bio-based local sustainable circular wear	101000632	IA	https://cordis.europa.eu/project/id/101000632	Sustainable Economy
HMCS	Handheld Molecular Contaminant Screener	854670	SME-1	https://cordis.europa.eu/project/id/854670	Food Quality/Fraud
HUB4NGI	A Collaborative Platform to Unlock the Value of Next Generation Internet Experimentation	732569	CSA	https://cordis.europa.eu/project/id/732569	NGI
Human City Platform	A Human City Platform for town and city councils to foster citizen engagement	711389	SME-2	https://cordis.europa.eu/project/id/711389	Smart City
I Go Slow	Blockchain based platform for bringing slow food ideas and family owned farming to cyberspace	868842	SME-1	https://cordis.europa.eu/project/id/868842	Digital Economy
ICARUS	Aviation-driven Data Value Chain for Diversified Global and Local Operations	780792	IA	https://cordis.europa.eu/project/id/780792	Big Data
ICN2020	ICN2020: Advancing ICN towards real-world deployment through research, innovative applications, and global scale experimentation	723014	RIA	https://cordis.europa.eu/project/id/723014	ICN
ICONET	New ICT infrastructure and reference architecture to support Operations in future PI Logistics NETworks	769119	RIA	https://cordis.europa.eu/project/id/769119	IoT
ICT-BIOCHAIN	ICT Tools in Efficient Biomass Supply Chains for Sustainable Chemical Production	792221	BBI-CSA	https://cordis.europa.eu/project/id/792221	IoT

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ID Ward	ID Ward: blockchain-based universal ID for privacy-preserving personalization	887317	SME-1	https://cordis.europa.eu/project/id/887317	Digital Society
I-Media-Cities	Innovative e-environment for Research on Cities and the Media	693559	IA	https://cordis.europa.eu/project/id/693559	Media
IMPULSE	Identity Management in PUbLic SErvices	101004459	RIA	https://cordis.europa.eu/project/id/101004459	AI
IN2DREAMS	INtelligent solutions 2ward the Development of Railway Energy and Asset Management Systems in Europe	777596	Shift2Rail-RIA	https://cordis.europa.eu/project/id/777596	Cloud/IoT/AI
INFINITECH	Tailored IoT & BigData Sandboxes and Testbeds for Smart, Autonomous and Personalized Services in the European Finance and Insurance Services Ecosystem	856632	IA	https://cordis.europa.eu/project/id/856632	Big Data
InnoRenew CoE	Renewable materials and healthy environments research and innovation centre of excellence	739574	SGA-CSA	https://cordis.europa.eu/project/id/739574	Renewable Materials and Healthy Environments
INSPECTO	INSPECTO: A disruptive portable device with an innovative Method for Pesticides and contaminants Detection in Food	808038	SME-1	https://cordis.europa.eu/project/id/808038	Digital Economy
INSPECTr	Intelligence Network and Secure Platform for Evidence Correlation and Transfer (INSPECTr)	833276	RIA	https://cordis.europa.eu/project/id/833276	Secure Societies
INSPIRE-5Gplus	INtelligent Security and PervasIve tRust for 5G and Beyond	871808	RIA	https://cordis.europa.eu/project/id/871808	5G
IntellIoT	Intelligent, distributed, human-centered and trustworthy IoT environments	957218	RIA	https://cordis.europa.eu/project/id/957218	IoT
INTERFACE	TSO-DSO-Consumer INTERFACE aRchitecture to provide innovative grid services for an efficient power system	824330	IA	https://cordis.europa.eu/project/id/824330	Smart Grid
INTERLACE	Interacting Decentralized Transactional and Ledger Architecture for Mutual Credit	754494	CSA	https://cordis.europa.eu/project/id/754494	FinTech
InteropEHRate	Interoperable EHRs at user edge	826106	RIA	https://cordis.europa.eu/project/id/826106	Healthcare
InterQ	Interlinked Process, Product and Data Quality framework for Zero-Defects Manufacturing	958357	IA	https://cordis.europa.eu/project/id/958357	Smart Factory
IoTCrawler	IoTCrawler	779852	RIA	https://cordis.europa.eu/project/id/779852	IoT
IoT-NGIN	Next Generation IoT as part of Next Generation Internet	957246	RIA	https://cordis.europa.eu/project/id/957246	IoT
iReceptor Plus	Architecture and tools for the query of antibody and T-cell receptor sequencing data repositories for enabling improved personalized medicine and immunotherapy	825821	RIA	https://cordis.europa.eu/project/id/825821	Healthcare
JUMP2Excel	Joint Universal activities for Mediterranean PV integration Excellence	810809	CSA	https://cordis.europa.eu/project/id/810809	Smart Grid

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KIOS CoE	KIOS Research and Innovation Centre of Excellence	739551	SGA-CSA	https://cordis.europa.eu/project/id/739551	Sustainable Economy
KONFIDO	Secure and Trusted Paradigm for Interoperable eHealth Services	727528	RIA	https://cordis.europa.eu/project/id/727528	Healthcare
KRAKEN	Brokerage and market platform for personal data	871473	IA	https://cordis.europa.eu/project/id/871473	Digital Information Management
LEDGER	decentraLizEd Data Governance for nExt geneRation internet	825268	RIA	https://cordis.europa.eu/project/id/825268	NGI
LightKone	Lightweight Computation for Networks at the Edge	732505	RIA	https://cordis.europa.eu/project/id/732505	Cloud
LOCARD	Lawful evidence collecting and continuity platform development	832735	RIA	https://cordis.europa.eu/project/id/832735	Secure Societies
LPS	For a full data privacy on the go.	886716	SME-1	https://cordis.europa.eu/project/id/886716	CyberSecurity
MANTIS	Cyber Physical System based Proactive Collaborative Maintenance	662189	RIA	https://cordis.europa.eu/project/id/662189	CyberSecurity
MANU-SQUARE	MANUfacturing ecoSystem of QUAlified Resources Exchange	761145	RIA	https://cordis.europa.eu/project/id/761145	Manufacturing
MARCO	Modern framework for Blockchain applications of corporate use	832468	SME-1	https://cordis.europa.eu/project/id/832468	Digital Economy
MARKET4.0	A Multi-Sided Business Platform for Plug and Produce Industrial Product Service Systems	822064	IA	https://cordis.europa.eu/project/id/822064	Manufacturing
MARSAL	Machine learning-based, networking and computing infrastructure resource management of 5G and beyond intelligent networks	101017171	RIA	https://cordis.europa.eu/project/id/101017171	5G
MAtchUP	MAximizing the UPscaling and replication potential of high level urban transformation strategies	774477	IA	https://cordis.europa.eu/project/id/774477	Smart City
MATRYCS	Modular Big Data Applications for Holistic Energy Services in Buildings	101000158	iA	https://cordis.europa.eu/project/id/101000158	Big Data
MDOT	Medical Device Obligations Taskforce	814654	IA	https://cordis.europa.eu/project/id/814654	Healthcare
MediaRoad	MediaRoad - European Media Ecosystem for Innovation	761412	CSA	https://cordis.europa.eu/project/id/761412	Media
MediaVerse	A universe of media assets and cocreation opportunities at your fingertips	957252	IA	https://cordis.europa.eu/project/id/957252	Media
MERWIS	Melanoma Early Warning System	878237	SME-1	https://cordis.europa.eu/project/id/878237	Healthcare
MH-MD	My Health - My Data	732907	RIA	https://cordis.europa.eu/project/id/732907	Healthcare
Minespider	Blockchain protocol for responsible mineral sourcing	835775	SME-1	https://cordis.europa.eu/project/id/835775	Digital Economy
MINESPIDER	Blockchain Protocol for Responsible Mineral Sourcing	946437	SME-2b	https://cordis.europa.eu/project/id/946437	Digital Economy
MOLIERE	MObiLIty sERvices Enhanced by GALILEO & Blockchain	101004275	IA	https://cordis.europa.eu/project/id/101004275	Mobility

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
M-Sec	Multi-layered Security technologies to ensure hyper connected smart cities with Blockchain, BigData, Cloud and IoT	814917	RIA	https://cordis.europa.eu/project/id/814917	Smart City
Navigato	CaaS - Car Charging as a Service for Smart Mobility	886231	SME-1	https://cordis.europa.eu/project/id/886231	Mobility
NECOS	Novel Enablers for Cloud Slicing	777067	RIA	https://cordis.europa.eu/project/id/777067	AI
NeMo	NeMo : Hyper-Network for electroMobility	713794	RIA	https://cordis.europa.eu/project/id/713794	Mobility
netCommons	network infrastructure as commons	688768	RIA	https://cordis.europa.eu/project/id/688768	Digital Society
NEWTRAL	First real-time fact-checking tool to fight against the fake news and disinformation	855556	SME-1	https://cordis.europa.eu/project/id/855556	Media
NewTREND	New integrated methodology and Tools for Retrofit design towards a next generation of ENergy efficient and sustainable buildings and Districts	680474	IA	https://cordis.europa.eu/project/id/680474	Circular Economy
NEXTLEAP	NEXTLEAP	688722	RIA	https://cordis.europa.eu/project/id/688722	Secure Societies
NGI FORWARD	NGI FORWARD	825652	CSA	https://cordis.europa.eu/project/id/825652	NGI
NGI MOVE	Ecosystem building blocks for the Next Generation Internet movement	780271	CSA	https://cordis.europa.eu/project/id/780271	NGI
NGI_TRUST	Partnership for innovative technological solutions to ensure privacy and enhance trust for the human-centric Internet	825618	RIA	https://cordis.europa.eu/project/id/825618	IoT
NGI0-Discovery	NGI Zero Discovery	825322	RIA	https://cordis.europa.eu/project/id/825322	NGI
NGI0-PET	NGI Zero - Privacy Enhancing Technologies	825310	RIA	https://cordis.europa.eu/project/id/825310	NGI
NGI-GO2S	NGI Research Goes To Standardization	957073	RIA	https://cordis.europa.eu/project/id/957073	Sustainable Economy
Ngrave	The World's Most Secure Cryptocurrency Safeguarding Solution	882699	SME-1	https://cordis.europa.eu/project/id/882699	Digital Society
NIMBLE	Collaboration Network for Industry, Manufacturing, Business and Logistics in Europe	723810	RIA	https://cordis.europa.eu/project/id/723810	Sustainable Economy
nIoVe	A Novel Adaptive Cybersecurity Framework for the Internet-of-Vehicles	833742	IA	https://cordis.europa.eu/project/id/833742	CyberSecurity
NRG-5	Enabling Smart Energy as a Service via 5G Mobile Network advances (NRG-5)	762013	RIA	https://cordis.europa.eu/project/id/762013	5G
OLYMPUS	Oblivious identitY Management for Private and User-friendly Services	786725	IA	https://cordis.europa.eu/project/id/786725	Digital Information Management
OntoChain	Trusted, traceable and transparent ontological knowledge on blockchain	957338	RIA	https://cordis.europa.eu/project/id/957338	Sustainable Economy
OPTIMAI	Optimizing Manufacturing Processes through Artificial Intelligence and Virtualization	958264	IA	https://cordis.europa.eu/project/id/958264	AI
P2B	Peers to blockchain	851033	CSA	https://cordis.europa.eu/project/id/851033	Digital Society

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
PANACEA	Protection and privAcy of hospital and health iNfrastructures with smArt Cyber sEcurity and cyber threat toolkit for dAta and people	826293	RIA	https://cordis.europa.eu/project/id/826293	CyberSecurity
PANORAMIX	Privacy and Accountability in Networks via Optimized Randomized Mix-nets	653497	IA	https://cordis.europa.eu/project/id/653497	Secure Societies
PARITY	Pro-sumer AwaRe, Transactive Markets for Valorization of Distributed flexibilITY enabled by Smart Energy Contracts	864319	IA	https://cordis.europa.eu/project/id/864319	Smart Grid
PatientDataChain	Blockchain approach to disrupt patient-provider medical records data exchange	886084	SME-1	https://cordis.europa.eu/project/id/886084	Healthcare
PCCE	Pricing carbon with a dedicated currency to empower economic agents	817104	SME-1	https://cordis.europa.eu/project/id/817104	Circular Economy
PCHP	Complete and optimize the PeachPie compiler technology to be fully compatible with the PHP language in order to increase th performance of up to 80% of the internet.	853706	SME-1	https://cordis.europa.eu/project/id/853706	ICT
PERISCOPE	Pan-European Response to the ImpactS of COVID-19 and future Pandemics and Epidemics	101016233	RIA	https://cordis.europa.eu/project/id/101010233	Healthcare
PharmaLedger	PharmaLedger	853992	IMI2-RIA	https://cordis.europa.eu/project/id/853992	Healthcare
PHOENIX	Electrical Power System's Shield against complex incidents and extensive cyber and privacy attacks	832989	IA	https://cordis.europa.eu/project/id/832989	CyberSecurity
Place to Plug	The collaborative blockchain backed platform bringing together disperse public and private EV charging point providers and drivers	868489	SME-1	https://cordis.europa.eu/project/id/868489	Smart Grid
PLANET	Progress towards Federated Logistics Through The Integration Of TEN-T into A Global Trade Network	860274	RIA	https://cordis.europa.eu/project/id/860274	Digital Economy
PlatOne	PLATform for Operation of distribution NEtworks	864300	IA	https://cordis.europa.eu/project/id/864300	Smart Grid
Pop-Machina	Collaborative production for the circular economy; a community approach	821479	IA	https://cordis.europa.eu/project/id/821479	Circular Economy
PoSeID-on	Protection and control of Secured Information by means of a privacy enhanced Dashboard	786713	IA	https://cordis.europa.eu/project/id/786713	CyberSecurity
Preemie	Personalized nutrition of low-birthweight infants	879228	SME-2	https://cordis.europa.eu/project/id/879228	Healthcare
PriMO-5G	Virtual Presence in Moving Objects through 5G	815191	RIA	https://cordis.europa.eu/project/id/815191	5G
PRIViLEDGE	Privacy-Enhancing Cryptography in Distributed Ledgers	780477	RIA	https://cordis.europa.eu/project/id/780477	CyberSecurity
PROCESS	PROviding Computing solutions for ExaScale ChallengeS	777533	RIA	https://cordis.europa.eu/project/id/777533	Big Data

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
Productive4.0	Electronics and ICT as enabler for digital industry and optimized supply chain management covering the entire product lifecycle	737459	IA	https://cordis.europa.eu/project/id/737459	Digital Economy
PROFILE	Data Analytics, Data Sources, and Architecture for Upgraded European Customs Risk Management	786748	RIA	https://cordis.europa.eu/project/id/786748	Big Data
PROTECTIVE	Proactive Risk Management through Improved Cyber Situational Awareness	700071	IA	https://cordis.europa.eu/project/id/700071	Secure Societies
ProtonSuite	The most secure collaboration suite in the world	848554	SME-2	https://cordis.europa.eu/project/id/848554	Secure Societies
ProtonSuite	The world's largest secure collaboration suite	791727	SME-1	https://cordis.europa.eu/project/id/791727	Secure Societies
PROVENANCE	Providing Verification Assistance for New Content	825227	IA	https://cordis.europa.eu/project/id/825227	Media
PTwist	An open platform for plastics lifecycle awareness, monetization, and sustainable innovation	780121	IA	https://cordis.europa.eu/project/id/780121	Circular Economy
PUZZLE	Towards a Sophisticated SIEM Marketplace for Blockchain-based Threat Intelligence and Security-as-aService	883540	IA	https://cordis.europa.eu/project/id/883540	CyberSecurity
QU4LITY	Digital Reality in Zero Defect Manufacturing	825030	IA	https://cordis.europa.eu/project/id/825030	Manufactoring
QualiChain	Decentralised Qualifications' Verification and Management for Learner Empowerment, Education Reengineering and Public Sector Transformation	822404	RIA	https://cordis.europa.eu/project/id/822404	Digital Society
QuardCard	Powered smart card with a biometric one time password system	757096	SME-2	https://cordis.europa.eu/project/id/757096	Secure Societies
QuidProQuo	True digitalisation of all transactional administration in the trade value chain through smart legal contracts	855945	SME-1	https://cordis.europa.eu/project/id/855945	Digital Economy
REFLOW	constRuctive mEtabolic processes For materiaL flOWs in urban and periurban environments across Europe	820937	IA	https://cordis.europa.eu/project/id/820937	Circular Economy
ResearchProof	An online digital logbook to protect and prove authorship, and to share scientific results	782642	SME-1	https://cordis.europa.eu/project/id/782642	Digital Economy
RESISTO	RESIlience enhancement and risk control platform for communication infraSTructure Operators	786409	IA	https://cordis.europa.eu/project/id/786409	CyberSecurity
RISE	Research Center on Interactive Media, Smart System and Emerging Technologies	739578	SGA-CSA	https://cordis.europa.eu/project/id/739578	Digital Society
Roksnet	e-Society Interconnections Software	816731	SME-1	https://cordis.europa.eu/project/id/816731	Digital Economy
SAFECARE	SAFEguard of Critical heAlth infrastructure	787002	IA	https://cordis.europa.eu/project/id/787002	CyberSecurity

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
SafeLog	Safe human-robot interaction in logistic applications for highly flexible warehouses	688117	RIA	https://cordis.europa.eu/project/id/688117	Digital Economy
SAGRIS	Sentinels-based Agriculture Information Service Component	782116	SME-1	https://cordis.europa.eu/project/id/782116	Space Data
SCOTT	Secure COnnected Trustable Things	737422	IA	https://cordis.europa.eu/project/id/737422	5G
SeafoodTrace	Intelligent Traceability Platform enabling full transparency in the Seafood supply chain	816070	SME-1	https://cordis.europa.eu/project/id/816070	IoT
SECREDAS	Cyber Security for Cross Domain Reliable Dependable Automated Systems	783119	ECSEL-RIA	https://cordis.europa.eu/project/id/783119	CyberSecurity
SecureCloud	Secure Big Data Processing in Untrusted Clouds	690111	RIA	https://cordis.europa.eu/project/id/690111	Cloud
SELIS	Towards a Shared European Logistics Intelligent Information Space	690588	RIA	https://cordis.europa.eu/project/id/690588	Digital Economy
SerIoT	Secure and Safe Internet of Things	780139	RIA	https://cordis.europa.eu/project/id/780139	IoT
SERUMS	Securing Medical Data in Smart Patient-Centric Healthcare Systems	826278	RIA	https://cordis.europa.eu/project/id/826278	Healthcare
SettleMint	The distributed Blockchain middleware that allows business worldwide to build business solutions with Blockchain technology	849969	SME-2	https://cordis.europa.eu/project/id/849969	Digital Economy
SHAR-Q	Storage capacity sharing over virtual neighbourhoods of energy ecosystems	731285	RIA	https://cordis.europa.eu/project/id/731285	Smart Grid
SHOGANAI	World´s first real-time solution for controlling airplane operating costs	806470	SME-2	https://cordis.europa.eu/project/id/806470	Digital Economy
Signa2.0	Signaturit	778550	SME-2	https://cordis.europa.eu/project/id/778550	Secure Societies
SIGNATURIT_2016	Signaturit - The smartest e-Signature solution to send and sign documents on the go	735095	SME-1	https://cordis.europa.eu/project/id/735095	Secure Societies
SKIN	Short supply chain Knowledge and Innovation Network	728055	CSA	https://cordis.europa.eu/project/id/728055	Digital Economy
SlideWiki	Large-scale pilots for collaborative OpenCourseWare authoring, multiplatform delivery and Learning Analytics	688095	IA	https://cordis.europa.eu/project/id/688095	Education
SlotMachine	A Privacy-Preserving Marketplace for Slot Management	890456	SESAR-RIA	https://cordis.europa.eu/project/id/890456	Air Traffic Management
SMark2.0	our next level product authentication, anti-counterfeiting and track and trace system	878178	SME-1	https://cordis.europa.eu/project/id/878178	Manufacturing
SmartCertificate	The trusted solution for issuing certified documents to the blockchain, checkable in just a click	837292	SME-1	https://cordis.europa.eu/project/id/837292	Secure Societies
SmartCOOLBOX	Smart modular passive cool boxes for sustainable logistics	827231	SME-1	https://cordis.europa.eu/project/id/827231	Secure Societies
SmartDegrees	First anti-fraud plataform for the registration and certification of diplomas and academic certificates based in blockchain	887349	SME-1	https://cordis.europa.eu/project/id/887349	Education

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
Smart-Trust	Secure Mobile ID for Trusted Smart Borders	778571	SME-2	https://cordis.europa.eu/project/id/778571	Secure Societies
SMASH	Smart Sharing	738441	SME-2	https://cordis.europa.eu/project/id/738441	Mobility
SMESEC	Protecting Small and Medium-sized Enterprises digital technology through an innovative cyber-SECurity framework	740787	IA	https://cordis.europa.eu/project/id/740787	CyberSecurity
SoBigData	SoBigData Research Infrastructure	654024	RIA	https://cordis.europa.eu/project/id/654024	Big Data
Social Digital Lab	Gamified and collaborative digital learning open-source platform with a blockchain-based system to facilitate crowdsourced learning and the implementation of personalized education strategies	817456	SME-1	https://cordis.europa.eu/project/id/817456	Education
SocialTruth	Open Distributed Digital Content Verification for Hyper-connected Sociality	825477	IA	https://cordis.europa.eu/project/id/825477	Media
SocKETs	Societal engagement with key enabling technologies	958277	CSA	https://cordis.europa.eu/project/id/958277	Digital Society
SOFIE	Secure Open Federation for Internet Everywhere	779984	RIA	https://cordis.europa.eu/project/id/779984	IoT
Solar Bank	Virtual Energy Trading IT System to couple photovoltaic production and electric vehicles charging.	815019	SME-1	https://cordis.europa.eu/project/id/815019	Digital Economy
SONNETS	SOcietal Needs aNalysis and Emerging Technologies in the public Sector	692868	CSA	https://cordis.europa.eu/project/id/692868	Digital Society
SOTER	cyberSecurity Optimization and Training for Enhanced Resilience in finance	833923	IA	https://cordis.europa.eu/project/id/833923	CyberSecurity
SpeakNGI.eu	Citizen-driven open consultation on Next Generation Internet	780125	CSA	https://cordis.europa.eu/project/id/780125	NGI
SpecEMS	Spectral Energy Management System for appliance-level analytics, control, and microgrid renewables trading.	856213	SME-1	https://cordis.europa.eu/project/id/856213	Digital Economy
SPHINX	A Universal Cyber Security Toolkit for Health-Care Industry	826183	RIA	https://cordis.europa.eu/project/id/826183	CyberSecurity
StandICT.eu 2023	ICT Standardization Observatory and Support Facility in Europe	951972	CSA	https://cordis.europa.eu/project/id/951972	ICT
Startup Lighthouse	Lighthouse: Lighting the way for European scale-ups	780738	IA	https://cordis.europa.eu/project/id/780738	Digital Economy
STMS	Smart Tyre Management System for Safer, Greener and More Economic Transport	728922	SME-1	https://cordis.europa.eu/project/id/728922	Manufacturing
STOP-IT	Strategic, Tactical, Operational Protection of water Infrastructure against cyber-physical Threats	740610	IA	https://cordis.europa.eu/project/id/740610	CyberSecurity
STOPTHEFRAUDINOLIVEO	Fighting counterfeiting in Olive Oil with blockchain - a working product has arrived	866703	SME-1	https://cordis.europa.eu/project/id/866703	Digital Economy

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
SUNFISH	SecUre iNFormation SHaring in federated heterogeneous private clouds	644666	RIA	https://cordis.europa.eu/project/id/644666	Cloud
SUPERCLOUD	User-centric management of security and dependability in clouds of clouds	643964	RIA	https://cordis.europa.eu/project/id/643964	Cloud
symbIoTe	Symbiosis of smart objects across IoT environments	688156	RIA	https://cordis.europa.eu/project/id/688156	IoT
Synx BIOS	Synx BIOS: The revolutionary webBIOS that fully enables the Internet of Things (IoT) through the Real Time Web	855028	SME-1	https://cordis.europa.eu/project/id/855028	IoT
System EYE	cutting-edge innovation to make your drive safer, collect and monetize automotive data	837186	SME-1	https://cordis.europa.eu/project/id/837186	IoT
TALENT SWARM 7D	Collaborative Work Environment for Project Lifecycle Management (PLM) of Complex Industrial Assets	816333	SME-1	https://cordis.europa.eu/project/id/816333	IoT
TCBL	Textile and Clothing Business Labs Transformative Business Models for the Textile Clothing Sector	646133	IA	https://cordis.europa.eu/project/id/646133	Supply Chain
TeraFlow	Secured autonomic traffic management for a Tera of SDN flows	101015857	RIA	https://cordis.europa.eu/project/id/101015857	5G
TERMINET	nexT gEneRation sMart INterconnectEd ioT	957406	RIA	https://cordis.europa.eu/project/id/957406	IoT
TeSLA	An Adaptive Trust-based eassesment System for Learning	688520	IA	https://cordis.europa.eu/project/id/688520	Digital Society
TITANIUM	Tools for the Investigation of Transactions in Underground Markets	740558	RIA	https://cordis.europa.eu/project/id/740558	CyberSecurity
TNT	Truth-not-Trust	881092	SME-2	https://cordis.europa.eu/project/id/881092	CyberSecurity
TOKEN	Transformative Impact Of BlocKchain tEchnologies iN Public Services	870603	RIA	https://cordis.europa.eu/project/id/870603	Digital Society
TRAPEZE	TRAPEZE - TRAnsparency, Privacy and security for European citiZEns	883464	IA	https://cordis.europa.eu/project/id/883464	Secure Societies
TRAXEN tracking	TRAXEN - Ultra low power, long life tracking sensor	836660	SME-1	https://cordis.europa.eu/project/id/836660	Supply Chain
TreBlo	Trusted and reliable content on future blockchains	957228	RIA	https://cordis.europa.eu/project/id/957228	NGI
TRESCA	Trustworthy, Reliable and Engaging Scientific Communication Approaches	872855	RIA	https://cordis.europa.eu/project/id/872855	Media
TrustEat	Building a trusty future food system by using blockchain tech	952600	CSA	https://cordis.europa.eu/project/id/952600	Food Quality/Fraud
TXpp	A technological solution for the 50 billion EUR VAT fraud problem	835042	SME-1	https://cordis.europa.eu/project/id/835042	Digital Economy
UNIFIED SCIENCE	The first academic social network creator that empowers researchers to collaborate, control privacy and increase their impact	854728	SME-1	https://cordis.europa.eu/project/id/854728	Digital Society

Horizon 2020 Project Acronym	Horizon 2020 Project Name	Horizon 2020 Project ID	Horizon 2020 Funding Scheme	CORDIS Project Factsheet	ISG PDL classification of Domain
UNIGEM	Next Generation 'Unified Guest Engagement System' for Smart Hospitality Services and Unparalleled Guest Loyalty	775371	SME-1	https://cordis.europa.eu/project/id/775371	Digital Society
VerifiedID	The online platform for Identity Verification that gives back personal data ownership to the citizens	835783	SME-1	https://cordis.europa.eu/project/id/835783	FinTech
VICINITY	Open virtual neighbourhood network to connect intelligent buildings and smart objects	688467	RIA	https://cordis.europa.eu/project/id/688467	IoT
VPP4ISLANDS	Virtual Power Plant for Interoperable and Smart isLANDS	957852	IA	https://cordis.europa.eu/project/id/957852	Smart Grid
WaMos	Wastewater Treatment Monitoring and Advisory System	868462	SME-1	https://cordis.europa.eu/project/id/868462	IoT
WeldGalaxy	Digital Dynamic Knowledge Platform for Welding in Manufacturing Industries	822106	IA	https://cordis.europa.eu/project/id/822106	Manufacturing
WeVerify	Wider and enhanced verification for you	825297	IA	https://cordis.europa.eu/project/id/825297	Media
WIDGET 3.0	Online tool to crowdfund NGO and Charity projects through the booming e-commerce market andbeyond!	711367	SME-2	https://cordis.europa.eu/project/id/711367	Digital Society
WORKERO	WORKERO - Connecting Space & Knowledge	827608	SME-1	https://cordis.europa.eu/project/id/827608	Smart City

Annex B:
Overview Horizon 2020 Research Topics

B.1 Horizon 2020 Research Topics

In the following an overview of all the H2020 Research Topics and thematic priorities under which the EU H2020 projects related to DLT and Blockchain from Table A.1 have been funded under is provided.

The different Horizon 2020 Research Topics are sorted by the H2020 Topic-ID in alphabetical order.

NOTE 1: The latest status update of the Horizon 2020 Research Topics related to the EU Horizon 2020 Projects information from CORDIS database is March 2021.

NOTE 2: The Thematic Priority information on the Horizon 2020 Research Topics has been retrieved from the Horizon Dashboard.

NOTE 3: No guarantee can be given for the completeness, correctness and topicality of the information and contents compiled here.

Table B.1: Overview H2020 Research Topics

Topic ID	H2020 Research Topic	Nr. of Projects
BBI.2017.S2	BBI.2017.S2 - Identify opportunities for ICT to increase the efficiency of biomass supply chains for the bio-based industry	1
CE-FNR-07-2020	CE-FNR-07-2020 - FOOD 2030 - Empowering cities as agents of food system transformation	1
CE-FNR-14-2020	CE-FNR-14-2020 - Innovative textiles - reinventing fashion	1
CE-SC5-03-2018	CE-SC5-03-2018 - Demonstrating systemic urban development for circular and regenerative cities	2
CIP-01-2016-2017	CIP-01-2016-2017 - Prevention, detection, response and mitigation of the combination of physical and cyber threats to the critical infrastructure of Europe.	5
CULT-COOP-11-2016-2017	CULT-COOP-11-2016-2017 - Understanding the transformation of European public administrations	1
DS-01-2014	DS-01-2014 - Privacy	1
DS-02-2016	DS-02-2016 - Cyber Security for SMEs, local public administration and Individuals	2
DS-03-2016	DS-03-2016 - Increasing digital security of health related data on a systemic le	1
DS-04-2015	DS-04-2015 - Information driven Cyber Security Management	1
DS-05-2016	DS-05-2016 - EU Cooperation and International Dialogues in Cybersecurity and Privacy Research and Innovation	1
DS-06-2017	DS-06-2017 - Cybersecurity PPP: Cryptography	2
DS-07-2017	DS-07-2017 - Cybersecurity PPP: Addressing Advanced Cyber Security Threats and Threat Actors	1
DS-08-2017	DS-08-2017 - Cybersecurity PPP: Privacy, Data Protection, Digital Identities	2
DT-FOF-08-2019	DT-FOF-08-2019 - Pilot lines for modular factories (IA 50%)	1
DT-FOF-11-2020	DT-FOF-11-2020 - Quality control in smart manufacturing (IA)	2
DT-GOVERNANCE-05-2018-2019-2020	DT-GOVERNANCE-05-2018-2019-2020 - New forms of delivering public goods and inclusive public services	2
DT-ICT-07-2018-2019	DT-ICT-07-2018-2019 - Digital Manufacturing Platforms for Connected Smart Factories	2
DT-ICT-11-2019	DT-ICT-11-2019 - Big data solutions for energy	2
DT-NMBP-02-2018	DT-NMBP-02-2018 - Open Innovation Test Beds for Safety Testing of Medical Technologies for Health (IA)	1
DT-NMBP-20-2018	DT-NMBP-20-2018 - A digital 'plug and produce' online equipment platform for manufacturing (IA)	2
DT-SPACE-01-EO-2018-2020	DT-SPACE-01-EO-2018-2020 - Copernicus market uptake	1
DT-SPACE-EGNSS-2-2019-2020	DT-SPACE-EGNSS-2-2019-2020 - EGNSS applications fostering digitization	1
DT-TDS-01-2019	DT-TDS-01-2019 - Smart and healthy living at home	1

Topic ID	H2020 Research Topic	Nr. of Projects
DT-TRANSFORMATIONS-02-2018-2019-2020	DT-TRANSFORMATIONS-02-2018-2019-2020 - Transformative impact of disruptive technologies in public services	4
ECSEL-01-2014	ECSEL-01-2014 - ECSEL Key Applications and Essential Technologies (RIA)	1
ECSEL-2016-2	ECSEL-2016-2 - ECSEL Key Applications and Essential Technologies (IA)	2
ECSEL-2017-2	ECSEL-2017-2 - RIA	1
ECSEL-2018-2	ECSEL-2018-2-RIA-Special-Topic - H2020-ECSEL-2018-2-RIA-Special-Topic	1
EE-20-2017	EE-20-2017 - Bringing to market more energy efficient and integrated data centres	1
EE-23-2017	EE-23-2017 - Innovative financing schemes	1
EeB-05-2015	EeB-05-2015 - Innovative design tools for refurbishing of buildings at district level	1
EIC-FTI-2018-2020	EIC-FTI-2018-2020 - Fast Track to Innovation (FTI)	3
EIC-SMEInst-2018-2020	EIC-SMEInst-2018-2020 - SME instrument	68
EINFRA-21-2017	EINFRA-21-2017 - Platform-driven e-infrastructure innovation	1
EUB-01-2017	EUB-01-2017 - Cloud Computing	1
EUB-1-2015	EUB-1-2015 - Cloud Computing, including security aspects	1
EUJ-01-2018	EUJ-01-2018 - Advanced technologies (Security/Cloud/IoT/BigData) for a hyper-connected society in the context of Smart City	1
EUJ-03-2016	EUJ-03-2016 - Experimental testbeds on Information-Centric Networking	1
EUK-01-2018	EUK-01-2018 - Cloud, IoT and AI technologies	1
EUK-02-2018	EUK-02-2018 - 5G	1
EURO-6-2015	EURO-6-2015 - Meeting new societal needs by using emerging technologies in the public sector	1
FETOPEN-01-2018-2019-2020	FETOPEN-01-2018-2019-2020 - FET-Open Challenging Current Thinking	1
FETOPEN-04-2016-2017	FETOPEN-04-2016-2017 - FET Innovation Launchpad	1
FOF-11-2016	FOF-11-2016 - Digital automation	3
FOF-12-2017	FOF-12-2017 - ICT Innovation for Manufacturing SMEs (I4MS)	1
GV-8-2015	GV-8-2015 - Electric vehicles' enhanced performance and integration into the transport system and the grid	1
H2020-TWINN-2015	H2020-TWINN-2015 - Twinning	2
ICT-01-2016	ICT-01-2016 - Smart Cyber-Physical Systems	1
ICT-06-2016	ICT-06-2016 - Cloud Computing	1
ICT-07-2014	ICT-07-2014 - Advanced Cloud Infrastructures and Services	3
ICT-07-2017	ICT-07-2017 - 5G PPP Research and Validation of critical technologies and systems	2
ICT-08-2017	ICT-08-2017 - 5G PPP Convergent Technologies	1
ICT-08-2019	ICT-08-2019 - Security and resilience for collaborative manufacturing environments	1
ICT-10-2015	ICT-10-2015 - Collective Awareness Platforms for Sustainability and Social Innovation	2
ICT-11-2017	ICT-11-2017 - Collective Awareness Platforms for Sustainability and Social Innovation	2
ICT-11-2018-2019	ICT-11-2018-2019 - HPC and Big Data enabled Large-scale Test-beds and Applications	1
ICT-12-2016	ICT-12-2016 - Net Innovation Initiative	1
ICT-13-2016	ICT-13-2016 - Future Internet Experimentation - Building a European experimental Infrastructure	1
ICT-13-2018-2019	ICT-13-2018-2019 - Supporting the emergence of data markets and the data economy	2
ICT-14-2016-2017	ICT-14-2016-2017 - Big Data PPP: cross-sectorial and cross-lingual data integration and experimentation	3
ICT-15-2016-2017	ICT-15-2016-2017 - Big Data PPP: Large Scale Pilot actions in sectors best benefitting from data-driven innovation	1
ICT-18-2016	ICT-18-2016 - Big data PPP: privacy-preserving big data technologies	1
ICT-19-2015	ICT-19-2015 - Technologies for creative industries, social media and convergence	1
ICT-19-2017	ICT-19-2017 - Media and content convergence	3
ICT-20-2015	ICT-20-2015 - Technologies for better human learning and teaching	2
ICT-20-2019-2020	ICT-20-2019-2020 - 5G Long Term Evolution	2
ICT-23-2019	ICT-23-2019 - EU-Taiwan 5G collaboration	1

Topic ID	H2020 Research Topic	Nr. of Projects
ICT-24-2015	ICT-24-2015 - Robotics	1
ICT-24-2018-2019	ICT-24-2018-2019 - Next Generation Internet - An Open Internet Initiative	4
ICT-28-2018	ICT-28-2018 - Future Hyper-connected Sociality	5
ICT-30-2015	ICT-30-2015 - Internet of Things and Platforms for Connected Smart Objects	4
ICT-30-2019-2020	ICT-30-2019-2020 - An empowering, inclusive Next Generation Internet	1
ICT-32-2017	ICT-32-2017 - Startup Europe for Growth and Innovation Radar	2
ICT-33-2019	ICT-33-2019 - Startup Europe for Growth and Innovation Radar	1
ICT-35-2018	ICT-35-2018 - Fintech: Support to experimentation frameworks and regulatory compliance	1
ICT-41-2017	ICT-41-2017 - Next Generation Internet	2
ICT-44-2020	ICT-44-2020 - Next Generation Media	2
ICT-45-2020	ICT-45-2020 - Reinforcing European presence in international ICT standardization: Standardization Observatory and Support Facility	1
ICT-52-2020	ICT-52-2020 - 5G PPP - Smart Connectivity beyond 5G	3
ICT-54-2020	ICT-54-2020 - Blockchain for the Next Generation Internet	3
ICT-56-2020	ICT-56-2020 - Next Generation Internet of Things	4
IMI2-2018-15-02	IMI2-2018-15-02 - Blockchain Enabled Healthcare	1
INFRADEV-4-2014-2015	INFRADEV-4-2014-2015 - Implementation and operation of cross-cutting services and solutions for clusters of ESFRI and other relevant research infrastructure initiatives	1
INFRAIA-1-2014-2015	INFRAIA-1-2014-2015 - Integrating and opening existing national and regional research infrastructures of European interest	2
INNOSUP-01-2018-2020	INNOSUP-01-2018-2020 - Cluster facilitated projects for new industrial value chains	1
INNOSUP-02-2016	INNOSUP-02-2016 - European SME innovation Associate - pilot	1
INNOSUP-03-2018	INNOSUP-03-2018 - Blockchain and distributed ledger technologies for SMEs	3
INNOSUP-05-2018-2020	INNOSUP-05-2018-2020 - Peer learning of innovation agencies	2
INNOSUP-2-2014	INNOSUP-2-2014 - IPR helpdesk	1
INSO-10-2015	INSO-10-2015 - SME business model innovation	2
INSO-10-2015-1	INSO-10-2015-1 - SME business model innovation	1
IoT-03-2017	IoT-03-2017 - R&I on IoT integration and platforms	5
LCE-01-2016-2017	LCE-01-2016-2017 - Next generation innovative technologies enabling smart grids, storage and energy system integration with increasing share of renewables: distribution network	5
LC-SC3-B4E-6-2020	LC-SC3-B4E-6-2020 - Big data for buildings	1
LC-SC3-EC-3-2020	LC-SC3-EC-3-2020 - Consumer engagement and demand response	1
LC-SC3-EE-10-2018-2019-2020	LC-SC3-EE-10-2018-2019-2020 - Mainstreaming energy efficiency finance	1
LC-SC3-ES-1-2019	LC-SC3-ES-1-2019 - Flexibility and retail market options for the distribution grid	4
LC-SC3-ES-4-2018-2020	LC-SC3-ES-4-2018-2020 - Decarbonising energy systems of geographical Islands	1
LC-SC3-ES-5-2018-2020	LC-SC3-ES-5-2018-2020 - TSO - DSO - Consumer: Large-scale demonstrations of innovative grid services through demand response, storage and small-scale (RES) generation	2
MG-2-9-2019	MG-2-9-2019 - InCo Flagship on Integrated multimodal, low-emission freight transport systems and logistics	2
MG-5-2-2017	MG-5-2-2017 - Innovative ICT solutions for future logistics operations	1
MG-5-4-2017	MG-5-4-2017 - Potential of the Physical Internet	1
MG-6.3-2015	MG-6.3-2015 - Common communication and navigation platforms for pan-European logistics applications	2
NMBP-22-2017	NMBP-22-2017 - Business models and industrial strategies supporting novel supply chains for innovative product-services	1
NMBP-38-2020	NMBP-38-2020 - Citizens and industrial technologies (CSA)	1
NMP-35-2014	NMP-35-2014 - Business models with new supply chains for sustainable customer-driven small series production	1
PHC-25-2015	PHC-25-2015 - Advanced ICT systems and services for integrated care	1
REFLECTIVE-6-2015	REFLECTIVE-6-2015 - Innovation ecosystems of digital cultural assets	1
RUR-10-2016-2017	RUR-10-2016-2017 - Thematic Networks compiling knowledge ready for practice	1
S2R-OC-IP3-01-2017	S2R-OC-IP3-01-2017 - Smart metering and asset management of railway systems	1
S2R-OC-IPX-03-2018	S2R-OC-IPX-03-2018 - Innovative/breakthrough mobility concepts (with rail as backbone)	1

Topic ID	H2020 Research Topic	Nr. of Projects
SC1-BHC-05-2018	SC1-BHC-05-2018 - International flagship collaboration with Canada for human data storage, integration and sharing to enable personalized medicine approaches	2
SC1-DTH-08-2018	SC1-DTH-08-2018 - Prototyping a European interoperable Electronic Health Record (EHR) exchange	1
SC1-PHE-CORONAVIRUS-2020-2C	SC1-PHE-CORONAVIRUS-2020-2C - Behavioural, social and economic impacts of the outbreak response	1
SCC-1-2016-2017	SCC-1-2016-2017 - Smart Cities and Communities lighthouse projects	1
SEC-12-FCT-2016-2017	SEC-12-FCT-2016-2017 - Technologies for prevention, investigation, and mitigation in the context of fight against crime and terrorism	2
SEC-17-BES-2017	SEC-17-BES-2017 - Architectures and organizations, big data and data analytics for customs risk management of the international goods supply chain trade movements	1
SESAR-ER4-04-2019	SESAR-ER4-04-2019 - Digital Information Management	1
SESAR-ER4-27-2019	SESAR-ER4-27-2019 - Digitalisation of the ATM Architecture	1
SFS-08-2018-2019	SFS-08-2018-2019 - Improving animal welfare	1
SFS-37-2019	SFS-37-2019 - Integrated approaches to food safety controls across the food chain	1
SFS-45-2016	SFS-45-2016 - Increase overall transparency of processed agri-food products	1
SMEInst-01-2016-2017	SMEInst-01-2016-2017 - Open Disruptive Innovation Scheme	7
SMEInst-04-2016-2017	SMEInst-04-2016-2017 - Engaging SMEs in space research and development	1
SMEInst-06-2016-2017	SMEInst-06-2016-2017 - Accelerating market introduction of ICT solutions for Health, Well-Being and Ageing Well	1
SMEInst-07-2016-2017	SMEInst-07-2016-2017 - Stimulating the innovation potential of SMEs for sustainable and competitive agriculture, forestry, agri-food and bio-based sectors	1
SMEInst-09-2016-2017	SMEInst-09-2016-2017 - Stimulating the innovation potential of SMEs for a low carbon and efficient energy system	1
SMEInst-10-2016-2017	SMEInst-10-2016-2017 - Small business innovation research for Transport and Smart Cities Mobility	2
SMEInst-13-2016-2017	SMEInst-13-2016-2017 - Engaging SMEs in security research and development	5
SU-BES02-2018-2019-2020	SU-BES02-2018-2019-2020 - Technologies to enhance border and external security	1
SU-DRS02-2018-2019-2020	SU-DRS02-2018-2019-2020 - Technologies for first responders	1
SU-DS03-2019-2020	SU-DS03-2019-2020 - Digital Security and privacy for citizens and Small and Medium Enterprises and Micro Enterprises	3
SU-DS04-2018-2020	SU-DS04-2018-2020 - Cybersecurity in the Electrical Power and Energy System (EPES): an armour against cyber and privacy attacks and data breaches	1
SU-DS05-2018-2019	SU-DS05-2018-2019 - Digital security, privacy, data protection and accountability in critical sectors	2
SU-FCT02-2018-2019-2020	SU-FCT02-2018-2019-2020 - Technologies to enhance the fight against crime and terrorism	3
SU-ICT-01-2018	SU-ICT-01-2018 - Dynamic countering of cyber-attacks	2
SU-ICT-02-2020	SU-ICT-02-2020 - Building blocks for resilience in evolving ICT systems	2
SU-ICT-03-2018	SU-ICT-03-2018 - Establishing and operating a pilot for a Cybersecurity Competence Network to develop and implement a common Cybersecurity Research & Innovation Roadmap	3
SU-INFRA01-2018-2019-2020	SU-INFRA01-2018-2019-2020 - Prevention, detection, response and mitigation of combined physical and cyber threats to critical infrastructure in Europe	1
SU-TDS-02-2018	SU-TDS-02-2018 - Toolkit for assessing and reducing cyber risks in hospitals and care centres to protect privacy/data/infrastructures	4
SwafS-19-2018-2019-2020	SwafS-19-2018-2019-2020 - Taking stock and re-examining the role of science communication	1
WIDESPREAD-01-2016-2017	WIDESPREAD-01-2016-2017 - Teaming Phase 2	3
WIDESPREAD-05-2017	WIDESPREAD-05-2017 - Twinning	1
WIDESPREAD-05-2020	WIDESPREAD-05-2020 - Twinning	1

Topic ID	H2020 Research Topic	Nr. of Projects
WIDESPREAD-06-2020	WIDESPREAD-06-2020 - ERA Chairs	1
WIDESPREAD-2-2014	WIDESPREAD-2-2014 - ERA Chairs	1

Annex C:
Overview International Research Programmes and Topics

C.1 Research Programmes US

In Table C.1 an overview of DLT and blockchain related research projects funded under the United States (US) National Science Foundation (NSF) programmes is provided.

NOTE 1: The latest status update of the project information is February 2021.

NOTE 2: The classification of the projects Core Technologies and or Vertical Domain is based on an ISG PDL assessment and therefore might differ from the actual projects core technologies or vertical domain.

NOTE 3: No guarantee can be given for the completeness, correctness and topicality of the information and contents compiled here.

Table C.1: US NSF Projects related to DLT and Blockchain as of February 2021

US NSF Award Project Name	US NSF Award #	US NSF Program	US NSF Project Starting Date	US NSF Award Factsheet	ISG PDL classification Core Technologies and/or Vertical Domain
Measurement, Analysis, and Novel Applications of Blockchains	1651938	CAREER	2017-02-15	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1651938	Core Technology Applications
Making Blockchains Scale Privately and Reliably	1719196	SaTC	2017-08-15	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1719196	Core Technology Scalability
Towards Privacy and Availability of Inter-blockchain Communication	1846316	CAREER	2019-02-15	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1846316	Core Technology InterLedger
Reducing Claims Denials in Healthcare Through Blockchain and Machine Learning	1914203	STTR (Phase I)	2019-07-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1914203	Vertical Domain Healthcare
The Use of Distributed Ledger Technology in Climate Governance	1932220	NSF 2026 Fund	2019-08-15	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1932220	Vertical Domain Climate
Toward Enforceable Data Usage Control in Cloud-based IoT Systems	1916902	SaTC	2019-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1916902	Vertical Domain IoT

US NSF Award Project Name	US NSF Award #	US NSF Program	US NSF Project Starting Date	US NSF Award Factsheet	ISG PDL classification Core Technologies and/or Vertical Domain
Blockchain Architectures for Resource-Constrained Devices	1937357	SaTC	2019-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1937357	Vertical Domain IoT, Healthcare
Network Architecture and Routing Protocols for Payment Channel Networks	1910676	NeTS	2019-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1910676	Core Technology Payment Channel Networks
Blockchain-Backed Financial Services for Refugees and Migrants	1946229	SBIR (Phase I)	2020-02-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1946229	Vertical Domain FinTech
Blockchain-backed System that Establishes the Provenance of Digital Images and Videos	2011744	I-Corps	2020-06-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2011744	Vertical Domain Media
Distributed Ledger Technology as a Security and Tracking Mechanism for Online Gaming	2038357	I-Corps	2020-07-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2038357	Vertical Domain Media
Automated Support for Writing High-Assurance Smart Contracts	1801369	SaTC	2020-07-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=1801369	Core Technology Smart Contract
Blockchain-Enabled Machine Learning on Confidential Data	2026404	STTR (Phase II)	2020-08-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2026404	Vertical Domain Digital Information Management
Secure blockchain communication for federal benefit assessments during COVID-19	2031813	SBIR (Phase I)	2020-09-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2031813	Vertical Domain Healthcare
Self-Adaptive Cyber Risk Management via Machine to Machine Economy Supported by Blockchain and Smart Contracts Technology	2000792	SaTC	2020-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2000792	Vertical Domain Digital Economy
Blockchain-based Mechanisms for Timed Data Release and Timed Transactions	2020071	SaTC	2020-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2020071	Vertical Domain Digital Information Management

US NSF Award Project Name	US NSF Award #	US NSF Program	US NSF Project Starting Date	US NSF Award Factsheet	ISG PDL classification Core Technologies and/or Vertical Domain
Self-Adaptive Cyber Risk Management via Machine to Machine Economy Supported by Blockchain and Smart Contracts Technology	2000792	SaTC	2020-10-01	https://www.nsf.gov/awardsearch/showAward?AWD_ID=2000792	Vertical Domain CyberSecurity

Annex D:
Change History

Date	Version	Information about changes
June 2021	0.0.5	Inserted mindmaps, inserted RP2021 action requests, inserted clause 7 text.
July 2021	0.0.7	Final editorial corrections

History

Document history		
V1.1.1	September 2021	Publication